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KOTA (RAJASTHAN)
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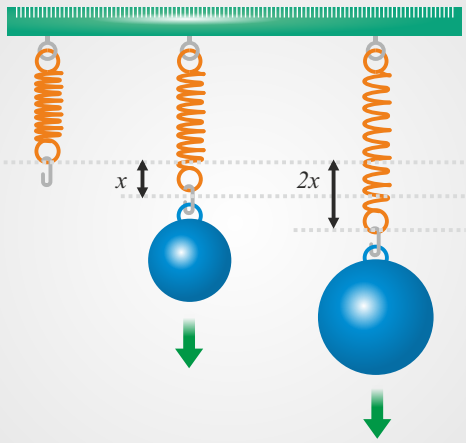
PHYSICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Exercise

Elasticity, Thermal Expansion, Calorimetry & Heat Transfer

(English Medium)

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Elasticity

1. The maximum load a wire can withstand without breaking, when its length is reduced to half of its original length, will
(A) be double. (B) be half. (C) be four times. (D) remain same.
EH0042

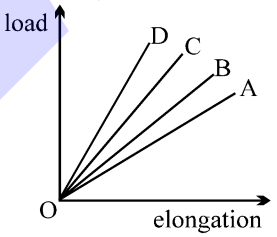
2. Two blocks of masses 1 kg and 2 kg are connected by a metal wire going over a smooth pulley. The breaking stress of metal is $\frac{40}{3\pi}\times10^6 N/m^{-2}$. What should be the minimum radius of wire used, if it should not be broken? ($g = 10ms^{-2}$) [ENG-2011]
(A) 0.5 mm (B) 1 mm (C) 1.5 mm (D) 2 mm **EH0237**

3. A wire can sustain the weight of 20 kg before breaking. If the wire is cut two equal parts, each part can sustain a weight of
(A) 10 kg (B) 20 kg (C) 40 kg (D) 80 kg **EH0238**

4. The elastic limit of steel is 8×10^8 N/m^2 and its Young's modulus 2×10^{11} N/m^2. Find the maximum elongation of a half-meter steel wire that can be given without exceeding the elastic limit.
(A) 2 mm (B) 4 mm (C) 1 mm (D) 16 mm **EH0239**

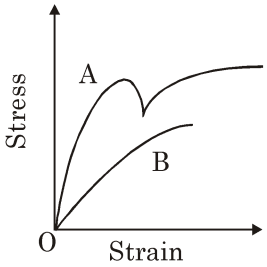
5. A lift is tied with thick iron wire and its mass is 1000kg. If the maximum acceleration of the lift is 1.2 ms^{-2} and the maximum stress of the wire is 1.4×10^8 Nm^{-2} what should be the minimum diameter of the wire?
(A) 10^{-2} m (B) 10^{-4} m (C) 10^{-6}m (D) 0.5×10^{-2}m
EH0240

6. The load versus strain graph for four wires of the same material is shown in the figure. The thickest wire is represented by the line :-



(A) OB (B) OA (C) OD (D) OC **EH0049**

7. The diagram show stress versus strain curve for the materials A and B . From the curves we infer that



(A) A is brittle but B is ductile (B) A is ductile and B is brittle
(C) both A and B are ductile (D) both A and B are brittle **EH0241**

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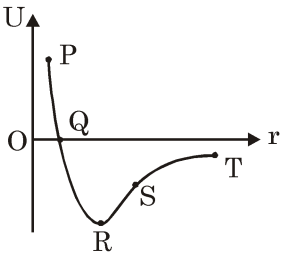
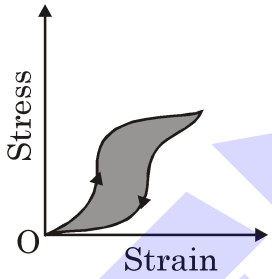
8. The point of maximum and minimum attraction in the curve between potential energy (U) and distance (r) of a diatomic molecules are respectively



(A) S and R (B) T and R (C) R and S (D) S and T **EH0242**

9. The diagram shows a stress-strain graph for a rubber band. Consider of the following statements



I. It will be easier to compress this rubber than expand it
II. Rubber does not return to its original length after it is stretched
III. The rubber band will get heated if it is stretched and released
Which of these can be deduced from the graph

(A) III only (B) II and III (C) I and III (D) I only **EH0243**

10. One end of a nylon rope of length 4.5 m and diameter 6 mm is fixed to a tree-limb. A monkey weighing 100 N jumps to catch the free end and stays there. Find the corresponding change in the diameter. Young's modulus of nylon = 4.8×10^{11} N/m^2 and Poisson's ratio of nylon = 0.2.

(A) 8.8×10^{-9} m (B) 4.4×10^{-9} m
(C) 1.76×10^{-8} m (D) 2.2×10^{-9} m. **EH0244**

11. Two wires of equal cross-section but one made of steel and the other of copper, are joined end to end. When the combination is kept under tension, the elongations in the two wires are found to be equal. Find the ratio of the lengths of the two wires. Young's modulus of steel = 2.0×10^{11} N/m^2 and that of copper = 1.1×10^{11} N/m^2.

(A) 2 : 1 (B) 20 : 11 (C) 1 : 1 (D) 11 : 20 **EH0245**

12. A wire elongates by 1.0 mm when load W is hanged from it. If this wire goes over a pulley and two weights W each are hung at the two ends, the elongation of the wire will be

(A) 0.5 mm (B) 1.0 mm (C) 2.0 mm (D) 4.0 mm **EH0246**

13. The length of a metal wire is ℓ_1 when the tension in it is T_1 and is ℓ_2 when the tension is T_2. The natural length of the wire is

(A) $\frac{\ell_1 + \ell_2}{2}$ (B) $\sqrt{\ell_1 \ell_2}$ (C) $\frac{\ell_1 T_2 - \ell_2 T_1}{T_2 - T_1}$ (D) $\frac{\ell_1 T_2 + \ell_2 T_1}{T_2 + T_1}$

EH0247

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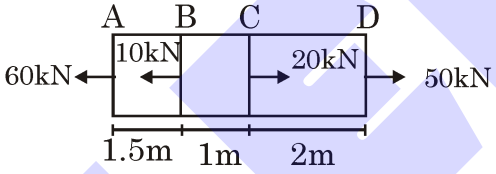
14. A heavy uniform rod is hanging verically from a fixed support. It is stretched by the its own weight. The diameter of the rod is
(A) smallest at the top and gradually increases down the rod
(B) largest at te the top and gradually decreases down the rod
(C) uniform every where
(D) maximum in the middle. **EH0248**

15. A copper wire of cross-sectional area 0.01 cm^2 is under a tension of 20 N. Find the decrease in the cross-sectional area. Young's modulus of copper = 1.1×10^{11} N/m^2 and Poisson's ratio = 0.32.
(A) 1.164×10^{-12} cm^2 (B) 1.164×10^{-6} cm^2 (C) 1.164×10^{-6} m^2 (D) 5.82×10^{-7} cm^2
EH0249

16. If x longitudinal strain is produced in a wire of Young's modulus y, then energy stored in the material of the wire per unit volume is
(A) yx^2 (B) $2yx^2$ (C) $\frac{1}{2}y^2x$ (D) $\frac{1}{2}yx^2$ **EH0250**

17. Wires A and B are made from the same material. A has twice the diameter and three times the length of B. If the elastic limits are not reached , when each is stretched by the same tension, the ratio of energy stored in A to that in B is
(A) 2 : 3 (B) 3 : 4 (C) 3 : 2 (D) 6 : 1 **EH0251**

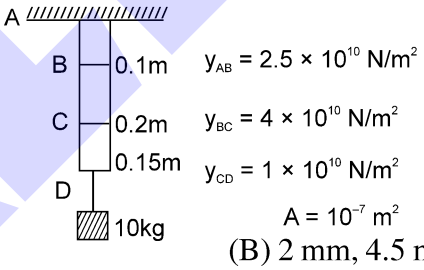
18. A Steel rod of cross-sectional area $1m^2$ is acted upon by forces as shown in the Fig. Determine the total elongation of the bar. Take $Y = 2.0 \times 10^{11} N/m^2$



(A) 1.3 μm (B) 1.7 μm (C) 0.9 μm (D) 1.2 μm **EH0252**

19. A wire is suspended vertically from one of its ends is stretched by attaching a weight of 200N to the lower end. The weight stretches the wire by 1mm. Then the elastic energy stored in the wire is
(A) 0.2J (B) 10 J (C) 20 J (D) 0.1 J **EH0253**

20. Find out the shift in point B, C and D respectively :-



(A) 4mm, 9 mm, 24 mm (B) 2 mm, 4.5 mm, 12 mm
(C) 6 mm, 13.5 mm, 36 mm (D) 3 mm, 6 mm, 18 mm **EH0254**

21. You do a very precise experiment to test the properties of a piece of wire. You stretch the wire by applying an outward force to each end of the wire and measure its change in length. If you then precisely double this force and the wire still remains elastic but is beyond proportionality limit, the change in length of the wire
(A) will exactly double (B) will just slightly more than double
(C) will just slightly less than double (D) will decrease by exactly a factor of two.
EH0255

22. A wire of area of cross-section 3.0 mm^2 and natural length 50 cm is fixed at one end and a mass of 2.1 kg is hung from the other end. Find the elastic potential energy stored in the wire in steady state. Young's modulus of the material of the wire = 1.9×10^{11} N/m^2. Take g = 10 m/s^2.
(A) 7.7×10^{-4} J (B) 1.9×10^{-4} erg (C) 3.8×10^{-4} J (D) 1.9×10^{-4} J
EH0256

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23. Overall changes in volume and radii of a uniform cylindrical steel wire are 0.2% and 0.002% respectively when subjected to some suitable force. Longitudinal tensile stress acting on the wire is :- ($Y = 2.0 \times 10^{11}$ Nm^{-2})

(A) 3.2×10^{9} Nm^{-2} (B) 3.2×10^{7} Nm^{-2} (C) 3.6×10^{9} Nm^{-2} (D) 3.9×10^{8} Nm^{-2}

EH0045

24. A solid sphere of radius R made of material of bulk modulus K is surrounded by a liquid in a cylindrical container. A massless piston of area A floats on the surface of the liquid. When a mass m is placed on the piston to compress the liquid, the fractional change in the radius of the sphere $\delta R/R$ is

(A) mg/AK (B) mg/3AK (C) mg/A (D) mg/3AR

EH0046

25. Find the volume density of the elastic deformation energy in fresh water at the depth of h = 1000 m.(The compressibility of water is 0.49 GPa^{-1})Hint : [P = ρgh]

(A) 24.5 kJ/m^3 (B) 47 kJ/m^3 (C) 11.25 kJ/m^3 (D) None of these

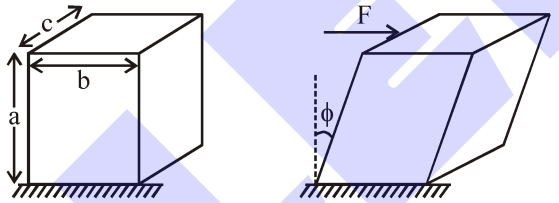
EH0257

26. A ball falling in a lake of depth 200 m shows 0.1% decrease in its volume at the bottom. What is the bulk modulus of the material of the ball :– Hint : [P = ρgh]

(A) 19.6×10^{8} N/m^2 (B) 19.6×10^{-10} N/m^2 (C) 19.6×10^{10} N/m^2 (D) 19.6×10^{-8} N/m^2

EH0258

27. A cuboidal block of sides a, b & c is fixed on ground. The top is pushed by a horizontal force F as shown. The angle ϕ by which the block deforms is : (η is modulus of rigidity)



(A) $\frac{F}{ab\eta}$ (B) $\frac{F}{ac\eta}$ (C) $\frac{F}{bc\eta}$ (D) $\frac{F}{\sqrt{b^2+c^2}\,\eta}$

EH0050

28. A steel plate of face-area 4 cm^2 and thickness 0.5 cm is fixed rigidly at the lower surface. A force of 9.6 N is applied parallel to the upper surface. Find the lateral displacement of the upper surface. Rigidity modulus of steel = 8×10^{10} N/m^2.

(A) 1.5×10^{-9} m (B) 1.5×10^{-8} m (C) 1.5×10^{-10} m (D) 1.5×10^{-7} m **EH0259**

29. A wire of 1m length and 4mm radius is clamped at upper end. The lower end is twisted by an angle of 30°. The angle of shear is (in degree)

(A) 0.120 (B) 1.20 (C) 120 (D) 0.0120 **EH0260**

30. A block of weight 15N slides on a horizontal table the co-efficient of sliding friction is 0.4. The area of the block in contact with the table is 0.05 m^2. The shearing stress will be

(A) 120 Nm^2 (B) 140 Nm^2 (C) 160 Nm^2 (D) 180 Nm^2 **EH0261**

31. When a rubber cord is stretched, the change in volume is negligible compared to the change in its linear dimension. Then Poisson's ratio for rubber is:

(A) infinite (B) Zero (C) 0.5 (D) –1 **EH0262**

32. The Poisson's ratio σ should satisfy the relation

(A) $-1 < \sigma < 0.5$ (B) $-0.5 < \sigma < 1.0$ (C) $0.5 < \sigma < 1.0$ (D) $-10 < \sigma < -0.5$

EH0263

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33. Young's modulus of a metal is 15×10^{11} pa. If its Poisson's ratio is 0.4. The bulk modulus of the metal in Pa is

(A) 25×10^{11} (B) 2.5×10^{11} (C) 250×10^{11} (D) 0.25×10^{11} **EH0264**

34. Y, K, η represent the Young's modulus,bulk modulus and rigidity modulus of a body respectively . If rigidity modulus is twice the bulk modulus, then

(A) Y = 5K/18 (B) Y = 5η/9 (C) Y = 9K/5 (D) Y = 18K/5 **EH0265**

35. For a given material the Young's modulus is 2.4 times that of its rigidity modulus. Its Poisson's ratio is:

(A) 2.4 (B) 1.2 (C) 0.4 (D) 0.2 **EH0266**

36. For a material $Y = 6.6 \times 10^{10}$ N/m^2 and bulk modulus $K = 11 \times 10^{10}$ N/m^2, then its Poisson's ratio is

(A) 0.8 (B) 0.35 (C) 0.7 (D) 0.4 **EH0267**

37. If the Poisson's ratio of a solid is 2/5, then the ratio of its young's modulus to the rigidity modulus is

(A) 5/4 (B) 7/15 (C) 14/9 (D) 14/5 **EH0268**

38. A material has Poisson's ratio 0.3. If a uniform rod of it suffers a longitudinal strain of 25×10^{-3}, then the percentage increase in its volume is

(A) 1% (B) 2% (C) 3% (D) 4% **EH0269**

Thermal expansion

39. A hole is made in a metal plate, when the temperature of metal is raised then the diameter of the hole will :

(A) Decrease

(B) Increase

(C) Remain same

(D) Answer depends upon the initial temperature of the metal **EH0065**

40. A steel tape gives correct measurement at 20°C. A piece of wood is being measured with the steel tape at 0°C. The reading is 25 cm on the tape, the real length of the given piece of wood must be:

(A) 25 cm (B) <25 cm (C) >25 cm (D) can not say

EH0067

41. A thin copper wire of length L increase in length by 1% when heated from temperature T_1 to T_2. What is the percentage change in area when a thin copper plate having dimensions $2L \times L$ is heated from T_1 to T_2?

(A) 1% (B) 2% (C) 3% (D) 4%

EH0069

42. Two rods one of aluminium of length ℓ_1 having coefficient of linear expansion α_a, and other steel of length ℓ_2 having coefficient of linear expansion α_s are joined end to end. The expansion in both the rods is same on variation of temperature. Then the value of $\frac{l_1}{l_1+l_2}$ is: **[JEE' (Scr) 2003]**

(A) $\frac{\alpha_s}{\alpha_a+\alpha_s}$ (B) $\frac{\alpha_s}{\alpha_a-\alpha_s}$ (C) $\frac{\alpha_a+\alpha_s}{\alpha_s}$ (D) None of these

EH0072

43. An open vessel is filled completely with oil which has same coefficient of volume expansion as that of the vessel. On heating both oil and vessel,

(A) the vessel can contain more volume and more mass of oil.

(B) the vessel can contain same volume and same mass of oil.

(C) the vessel can contain same volume but more mass of oil.

(D) the vessel can contain more volume but same mass of oil. **EH0073**

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44. A pendulum clock consists of an iron rod connected to a small, heavy bob. If it is designed to keep correct time at 20°C, how fast or slow will it go in 24 hours at 40° C ? Coefficient of linear expansion of iron= 1.2×10^{-5}/°C.

(A) 10.4 s, fast (B) 10.4 s, slow (C) 20.8 s, fast (D) 20.8 s, slow

EH0270

45 The coefficient of linear expansion of copper is 17×10^{-6} $(°C)^{-1}$. A copper statue is 93 m tall on the summer morning of temperature 25°C. What is maximum order of increase in magnitude of the height in statue (maximum temperature of day is 45°C)

(A) 0.1 mm (B) 1 mm (C) 10 mm (D) 100 mm

EH0110

46. A glass window is to be fit in an aluminium frame. The temperature on the working day is 40°C and the glass window measures exacly 20 cm × 30 cm. What should be the length and width of the aluminium frame so that there is no stress on the glass in winter even if the temperature drops to 0°C? Coefficients of linear expansion for glass and aluminium are 9.0×10^{-6}/C and 24×10^{-6}/°C respectively.

(A) 20.012 cm, 30.018 cm (B) 20.018 cm, 30.024 cm

(C) 20.006 cm, 30.012 cm (D) 20.015 cm, 30.020 cm **EH0271**

47. An iron rod and a copper rod lie side by side. As the temperature is changed, the difference in the lengths of the rods remains constant at a value of 10 cm. Find the length of iron rod at 0°C. Coefficients of linear expansion of iron and copper are 1.1×10^{-5}/°C and 1.7×10^{-5}/°C respectively.

(A) 28.3 cm (B) 18. 3 cm (C) 33.5 cm (D) 22.7 cm **EH0272**

48. A meter scale made of steal is calibrated at 20°C to give correct reading. Find the distance between 50 cm mark and 51 cm mark if scale is used at 10°C. Coefficient of linear expansion of steel is 1.1×10^{-5}/°C.

(A) 0.99989 cm (B) 1.00011 cm (C) 0.89989 cm (D) 1.10011 cm

EH0273

49. The length of each steel rail is 10 m in winter. The coefficient of linear expansion of steel is 0.000012 /°C and the temperature increases by 15° C in summer. The gap to be left between the rails

(A) 0.0018m (B) 0.0012m (C) 0.0022m (D) 0.05m **EH0274**

50. A meter scale made of steel reads accurately at 20°C. In a sensitive experiment, distances accurate upto 0.055 mm in 1 m are required. Find the range of temperature in which the experiment can be performed with this meter scale. Coefficient of linear expansion of steel = 11×10^{-6}/°C.

(A) 12°C to 28°C (B) 18°C to 22°C

(C) 15°C to 25°C (D) 10°C to 30°C **EH0275**

51. A pendulum clock gives correct time at 20° at a place where g = 9.800 m/s^2. The pendulum consists of a light steel rod connected to a heavy ball. It is taken to a different place where g = 9.788 m/s^2. At what temperature will it give correct time? Coefficient of linear expansion of steel = 12×10^{-6}/°C.

(A) 122°C (B) 71°C (C) –69°C (D) –82°C **EH0276**

52. A tungsten wire, 0.5 mm in diameter, is just stretched between two fixed points at a temperature of 40°C. Determine the tension in the wire when the temperature falls to 20°C. (coefficient of linear expansion of tungsten = 4.5×10^{-6}/°C; Young's modulus of tungsten = 3.45×10^{10}Nm^{-2})

(A) 0.6097N (B) 3.097N (C) 5.097N (D)7.097N **EH0277**

53. A rod of length 2m rests on smooth horizontal floor. If the rod is heated from 0°C to 20°C. Find the longitudinal strain developed? ($\alpha = 5 \times 10^{-5}$/°C)

(A) 10^{-3} (B) 2×10^{-3} (C) Zero (D) None

EH0066

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54. Two rods of different materials are clamped at their ends rigidly. When they are heated for the same rise in temperature, same thermal stresses are produced in them. If their Young's moduli are in the ratio x:y then ratio of coefficients of their linear expansion is

(A) x : y (B) y : x (C) $x^2 : y^2$ (D) $y^2 : x^2$ **EH0278**

55. The bulk modulus of copper is 1.4×10^{11} Pa and the coefficient of linear expansion is 1.7×10^{-5} $(C°)^{-1}$. What hydrostatic pressure is necessary to prevent a copper block from expanding when its temperature is increased from 20°C to 30°C?

(A) 6.0×10^5 Pa (B) 7.1×10^7 Pa (C) 5.2×10^6 Pa (D) 40 atm

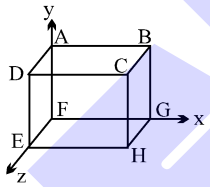
EH0068

56. The radius of a metal sphere at room temperature T is R, and the coefficient of linear expansion of the metal is α. The sphere is heated a little by a temperature ΔT so that its new temperature is $T + \Delta T$. The increase in the volume of the sphere is approximately

(A) $2\pi R\ \alpha\ \Delta T$ (B) $\pi R^2\ \alpha\ \Delta T$ (C) $4\pi R^3\ \alpha\ \Delta T$ /3 (D) $4\pi R^3\ \alpha\ \Delta T$

EH0064

57. A cuboid ABCDEFGH is anisotropic with $\alpha_x = 1 \times 10^{-5}$ /°C, $\alpha_y = 2 \times 10^{-5}$ /°C, $\alpha_z = 3 \times 10^{-5}$ /°C. Coefficient of superficial expansion of faces can be:



(A) $\beta_{ABCD} = 5 \times 10^{-5}$ /°C (B) $\beta_{BCGH} = 4 \times 10^{-5}$ /°C
(C) $\beta_{CDEH} = 3 \times 10^{-5}$ /°C (D) $\beta_{EFGH} = 2 \times 10^{-5}$ /°C

EH0070

58. A crystal has linear coefficients 0.00004 /°C, 0.00005 /°C, 0.00006 /°C. Coefficient of cubical expansion of the crystal is

(A) 0.000015 /°C (B) 0.00015 /°C (C) 0.00012 /°C (D) 0.00018 /°C

EH0279

59. The coefficient of apparent expansion of a liquid in a copper vessel is C and in a silver vessel is S. The coefficient of volume expansion of copper is γ_c. What is the coefficient of linear expansion of silver?

(A) $\frac{(C + \gamma_c + S)}{3}$ (B) $\frac{(C - \gamma_c + S)}{3}$ (C) $\frac{(C + \gamma_c - S)}{3}$ (D) $\frac{(C - \gamma_c - S)}{3}$

EH0071

60. The coefficients of thermal expansion of steel and a metal X are respectively 12×10^{-6} and 2×10^{-6} per°C. At 40°C, the side of a cube of metal X was measured using a steel vernier callipers. The reading was 100 mm. Assuming that the calibration of the vernier was done at 0°C, then the actual length of the side of the cube at 0°C will be

(A) > 100 mm (B) < 100 mm
(C) = 100 mm (D) data insufficient to conclude

EH0111

61. A rod of length 2m at 0°C and having expansion coefficient $\alpha = (3x + 2) \times 10^{-6}$ °C^{-1} where x is the distance (in cm) from one end of rod. The length of rod at 20°C is :

(A) 2.124 m (B) 3.24 m (C) 2.0120 m (D) 3.124 m

EH0113

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62. A glass vessel of volume 100 cm^3 is filled with mercury and is heated through 25°C. What volume of mercury will overflow? Coefficient of linear expansion of glass= 1.8×10^{-6}/°C. Coefficient of volumetric expansion of mercury = 1.8×10^{-4}/°C.
(A) 0.4365 cm^3 (B) 1.6 cm^3 (C) 1.74 cm^3 (D) 2.61 cm^3 **EH0280**

63. Two marks on a glass rod 10 cm apart are found to increase their distance by 0.06 mm when the rod is heated from 0 °C to 10 °C. A flask made of the same glass as that rod measures a volume of 1000 c.c at 0 °C. The volume it measures at 100 °C in c.c. is
(A) 1018 cc (B) 918 cc (C) 818 cc (D) 718 cc **EH0281**

64. An aluminium can of cylindrical shape contains 500 cm^3 of water. The area of the inner cross-section of the can is 125 cm^2. All measurements refer to 10°C/ Find the approximate rise in the water level if the temperature increases to 80°C. The coefficient of linear expansion of aluminium = 23×10^{-6}/°C and the average coefficient of volume expansion of water = 3.2×10^{-4}/°C respectively.
(A) 0.076 cm (B) 0.065 cm (C) 0.089 cm (D) 0.099 cm **EH0282**

65. A Celsius thermometer and a Fahrenheit thermometer are put in a hot bath. The reading on Fahrenheit thermometer is just 3 times the reading on Celsius thermometer. The temperature of the hot bath is
(A) 26.67 °C (B) 36.67°C (C) 46.67 °C (D) 56.67 °C **EH0283**

66. A Centigrade and a Fehrenheit thermometer are dipped in boiling water. The water temperature is lowered until the Fahrenheit thermometer registers 140°. What is the fall in temperature as registered by the Centigrade thermometer
(A) 30° (B) 40° (C) 60° (D) 80° **EH0284**

67. The reading of Centigrade thermometer coincides with that of Fahrenheit thermometer in a liquid. The temperature of the liquid is
(A) – 40 °C (B) 0 °C (C) 100 °C (D) 300 °C **EH0285**

68. On a hypothetical scale X, the ice point is 40° and the steam point is 120°. For another scale Y the ice point and steam points are –30° and 130° respectively. If X-reads 50° The reading of Y is
(A) –5° (B) –8° (C) –10° (D) –12° **EH0286**

Calorimetry

69. Equal amount of heat energy are transferred into equal mass of ethyl alcohol and water sample. The rise in temperature of water sample is 25°C. The temperature rise of ethyl alcohol will be.
(Specific heat of ethyl alcohol is one half of the specific heat of water).
(A) 12.5°C (B) 25°C
(C) 50°C (D) It depends on the rate of energy transfer.
EH0052

70. A block of mass 2.5 kg is heated to temperature of 500°C and placed on a large ice block. What is the maximum amount of ice that can melt (approx.). Specific heat for the body = 0.1 Cal/gm°C.
(A) 1 kg (B) 1.5 kg (C) 2 kg (D) 2.5 kg
EH0053

71. 10 gm of ice at 0°C is kept in a calorimeter of water equivalent 10 gm. How much heat should be supplied to the apparatus to evaporate the water thus formed? (Neglect loss of heat)
(A) 6200 cal (B) 7200 cal (C) 13600 cal (D) 8200 cal
EH0054

72. A continuous flow water heater (geyser) has an electrical power rating = 2 kW and efficienty of conversion of electrical power into heat = 80%. If water is flowing through the device at the rate of 100 cc/sec, and the inlet temperature is 10°C, the outlet temperature will be
(A) 12.2°C (B) 13.8°C (C) 20°C (D) 16.5°C
EH0055

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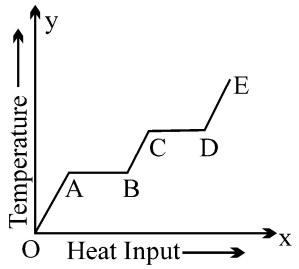
73. A solid material is supplied with heat at a constant rate. The temperature of material is changing with heat input as shown in the figure. What does slope DE represents?



(A) latent heat of liquid.
(B) latent heat of vapour.
(C) heat capacity of vapour.
(D) inverse of heat capacity of vapour.

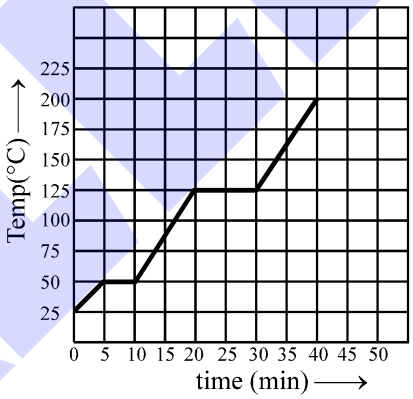
EH0056

74. The specific heat of a metal at low temperatures varies according to $S = aT^3$ where a is a constant and T is the absolute temperature. The heat energy needed to raise unit mass of the metal from T = 1 K to T = 2 K is :-

(A) 3 a (B) $\frac{15a}{4}$ (C) $\frac{2a}{3}$ (D) $\frac{12a}{5}$

EH0058

75. The graph shown in the figure represent change in the temperature of 5 kg of a substance as it abosrbs heat at a constant rate of 42 kJ min^{-1}. The latent heat of vapourazation of the substance is :



(A) 630 kJ kg^{-1} (B) 126 kJ kg^{-1} (C) 84 kJ kg^{-1} (D) 12.6 kJ kg^{-1}

EH0059

76. The density of a material A is 1500 kg/m^3 and that of another material B is 2000 kg/m^3. It is found that the heat capacity of 8 volumes of A is equal to heat capacity of 12 volumes of B. The ratio of specific heats of A and B will be

(A) 1 : 2 (B) 3 : 1 (C) 3 : 2 (D) 2 : 1

EH0060

77. 2 kg ice at – 20°C is mixed with 5 kg water at 20°C. Then final amount of water in the mixture would be; Given specific heat of ice = 0.5cal/g°C, specific heat of water = 1 cal/g°C, Latent heat of fusion of ice = 80 cal/g. **[JEE' (Scr) 2003]**

(A) 6 kg (B) 5 kg (C) 4 kg (D) 2 kg

EH0061

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78. Some steam at 100°C is passed into 1.1 kg of water contained in a calorimeter of water equivalent 0.02 kg at 15°C so that the temperature of the calorimeter and its contents rises to 80°C. What is the mass of steam condensing. (in kg)

(A) 0.130 (B) 0.065 (C) 0.260 (D) 0.135 **EH0062**

79. A block of ice with mass *m* falls into a lake. After impact, a mass of ice *m*/5 melts. Both the block of ice and the lake have a temperature of 0°C. If L represents the heat of fusion, the minimum distance the ice fell before striking the surface is

(A) $\frac{L}{5g}$ (B) $\frac{5L}{g}$ (C) $\frac{gL}{5m}$ (D) $\frac{mL}{5g}$ **EH0057**

80. A 5 g piece of ice at –20°C is put into 10 g of water at 30°C. Assuming that heat is exchanged only between the ice and the water, find the final temperature of the mixture. Specific heat capacity of ice = 2100 J kg^{-1} $°C^{-1}$, specific heat capacity of water = 4200 J kg^{-1} $°C^{-1}$ and latent heat of fusion of ice = 3.36×10^5 J kg^{-1}.

(A) –5°C (B) 0°C (C) 5°C (D) 10 °C **EH0287**

81. A lead ball at 30°C is dropped from a height of 6.2 km. The ball is heated due to the air resistance and it completely melts just before reaching the ground. The molten substance falls slowly on the ground. Calculate the latent heat of fusion of lead. Specific heat capacity of lead = 126 J $kg^{-1}°C^{-1}$ and melting point of lead = 330°C. Assume that any mechanical energy lost is used to heat the ball. Use g = 10 m s^{-2}.

(A) 6×10^3 J kg^{-1} (B) 1.2×10^4 J kg^{-1} (C) 1.8×10^4 J kg^{-1} (D) 2.4×10^4 J kg^{-1}

EH0288

82. 'n' number of liquids of masses m, 2m, 3m, 4m, ... having specific heats S, 2S,3S, 4S, ... are at temperatures t, 2t, 3t, 4t . . . . are mixed. The resultant temperature of mixture is

(A) $\frac{3n}{2n+1}t$ (B) $\frac{2n(n+1)}{3(2n+1)}t$ (C) $\frac{3n(n+1)}{2(2n+1)}t$ (D) $\frac{3n(n+1)}{(2n+1)}t$

EH0289

83. The temperature of equal masses of three different liquids A, B and C are 12°C, 19°C and 28°C respectively. The temperature when A and B are mixed is 16°C, and when B and C are mixed, it sis 23°C. What be the temperature when A and C are mixed?

(A) 20.3°C (B) 22.2°C (C) 23 °C (D) 24.8°C **EH0290**

84. Two rods of the same length, have radii in the ratio 3:4. Their densities are respectively 8000 and 9000 kg/m^3. Their specific heats are in the ratio of 2:3. When the same amount of heat is supplied to both, the changes in their lengths are in the ratio. (If their linear coefficients are in the ratio 5 : 6)

(A) 1 : 1 (B) 5 : 2 (C) 5 : 12 (D) 12 : 5 **EH0291**

85. 1 kg of ice at 0°C is mixed with 1 kg of steam at 100°C. What will be the composition of the system when thermal equilibrium is reached? latent heat of fusion of ice = 3.36×10^5 J/kg and latent heat of vaporization of water = 2.26×10^6 J/kg.

(A) 2000 g water (B) 665 g steam and 1335 g water

(C) 665 g ice and 1335 g water (D) 2000 g steam **EH0292**

86. A brick weighing 4.0 kg is dropped into a 1.0 m deep river from a height of 2.0 m. Assuming that 80% of the gravitational potential energy is finally converted into thermal energy, find this thermal energy in calorie.

(A) 21 cal (B) 21.5 cal (C) 23 cal (D) 24.5 cal **EH0293**

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87. A van of mass 1500 kg travelling at a speed of 54 km/h is stopped in 10 s. Assuming that all the mechanical energy lost appears as thermal energy in the brake mechanism, find the average rate of production of thermal energy in cal/s.

(A) 4000 cal/s (B) 3000 cal/s (C) 4200 cal/s (D) 3200 cal/s **EH0294**

88. On a winter day the temperature of the tap water is 20°C whereas the room temperature is 5°C. Water is stored in a tank of capacity 500 cm^3 for household use. If it were possible to use the heat liberated by the water to lift a 10 kg mass vertically, how high can it be lifted as the water comes to the room temperature ? Take g = 10 m/s^2.

(A) 255 m (B) 275 m (C) 295 m (D) 315 m **EH0295**

89. Two blocks of masses 10 kg and 20 kg moving at speeds of 10 m/s and 20 m/s respectively in opposite directions, approach each other and collide. If the collision is completely inelastic, find the thermal energy developed in the process.

(A) 2500 J (B) 3000 J (C) 4000 J (D) 4500 J **EH0296**

90. A ball is dropped on a floor from a height of 2.0 m. After the collision it rises up to a height of 1.5 m. Assume that 40% of the mechanical energy lost goes as thermal energy into the ball. Calculate the rise in the temperature of the ball in the collision. Specific heat capacity of the ball is 800 J/kg K.

(A) 1×10^{-3}°C (B) 1.5×10^{-3}°C (C) 2.5×10^{-3}°C (D) 3×10^{-3}°C

EH0297

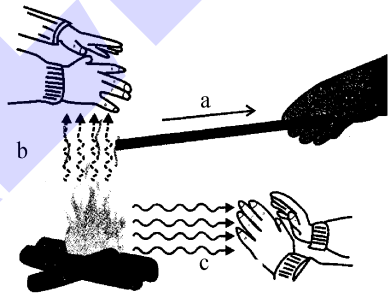
91. Ice at 0°C is added to 200 g of water initially at 70°C in a vacuum flask. When 50 g of ice has been added and has all melted the temperature of the flask and contents is 40°C. When a further 80g of ice has been added and has all metled, the temperature of the whole is 10°C. Calculate the specific latent heat of fusion of ice.[Take S_w =1 cal /gm °C.]

(A) 3.8×10^5 J/ kg (B) 1.2×10^5 J/ kg (C) 2.4×10^5 J/ kg (D) 3.0×10^5 J/ kg

EH0109

Conduction

92. Diagram shows a heat source 'S' and three position of heat recover (hand). The main mode of heat transfer is given as 'a', 'b' & 'c'. Choose the correct matching :-



(A) a – conduction ; b – convection ; c – radiation
(B) b – conduction ; a – convection ; c – radiation
(C) a – conduction ; c – convection ; b – radiation
(D) c – conduction ; b – radiation ; a – convection **EH0074**

93. One end of a metal rod is dipped in boiling water and the other is dipped in melting ice.

(A) All parts of the rod in thermal equilibrium with each other
(B) We can assign a temperature to the rod.
(C) We can assign a temperature to the rod after steady state is reached
(D) The thermal state of the rod does not change after steady state is reached. **EH0298**

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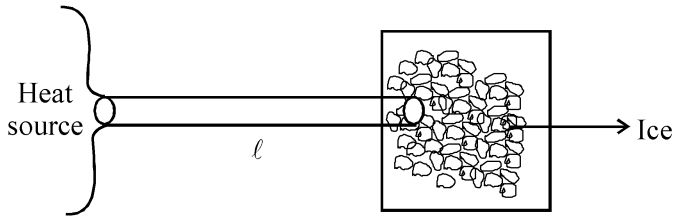
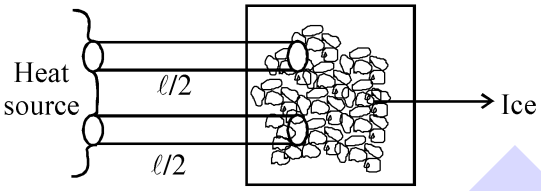
94. A rod of length 'ℓ' and cross-section 'A' is used to melt a piece of ice as shown.



Now if the rod broken into two equal parts and is arranged as shown.



Time taken to melt ice in second use becomes.

(A) Half (B) One-forth (C) Twice (D) Four times

EH0075

95. A liquid-nitrogen container is made of a 1 cm thick thermocol sheet having thermal conductivity 0.025 J/m–°C. Liquid nitrogen at 80K is kept in it. A total area of 0.80 m^2 is in contact with the liquid nitrogen. The atmospheric temperature is 300 K. Calculate the rate of heat flow from the atmosphere to the liquid nitrogen.

(A) 400 W (B) 440 W (C) 480 W (D) 500 W **EH0299**

96. One end of a 2.35m long and 2.0cm radius aluminium rod (K = 235 $W.m^{-1}K^{-1}$) is held at 20^0C. The other end of the rod is in contact with a block of ice at its melting point. The rate in $kg.s^{-1}$ at which ice melts is [Take latent heat of fusion for ice as $\frac{10}{3}\times10^5 J.kg^{-1}$]

(A) $48\pi \times 10^{-6}$ (B) $24\pi \times 10^{-6}$ (C) $2.4\pi \times 10^{-6}$ (D) $4.8\pi \times 10^{-6}$

EH0076

97. A pitcher with 1 mm thick porous walls contains 10 kg of water. Water comes to its outer surface and evaporates at the rate of 0.1 g/s. The surface area of the pitcher (one side) = 200 cm^2. The room temperature = 42°C, latent heat of vaporization = 2.27×10^6 J/kg, and the thermal conductivity of the porous walls = 0.80 J/m-s-°C. Calculate the approximate temperature of water in the pitcher when it attains a constant value.

(A) 28°C (B) 30°C (C) 32°C (D) 34°C **EH0300**

98. An icebox made of 1.5 cm thick styrofoam has dimensions 60 cm × 60 cm × 30 cm. It contains ice at 0°C and is kept in a room at 40°C. Find the rate at which the ice is melting. Latent heat of fusion of ice = 3.36×10^5 J kg^{-1} and thermal conductivity of styrofoam = 0.04 W m^{-1} $°C^{-1}$.

(A) 0.23 g s^{-1} (B) 0.46 g s^{-1} (C) 0.69 g s^{-1} (D) 0.92 g s^{-1} **EH0301**

99. The left end of a copper rod (length = 20 cm, area of cross-section = 0.20 cm^2) is maintained at 20°C and the right end is maintained at 80°C. neglecting any loss of heat through radiation, what is the temperature at a point 11 cm from the left end and the heat current through the rod respectively. Thermal conductivity of copper = 385 W/m-°C.

(A) 53°C & 2.31 J/s (B) 47°C & 2.31 J/s

(C) 53°C & 4.62 J/s (D) 47°C & 4.62 J/s **EH0302**

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100. Four rods of same material with different radii r and length ℓ are used to connect two reservoirs of heat at different temperatures. Which one will conduct most heat ?

(A) r = 2cm, ℓ = 0.5m (B) r = 2cm, ℓ = 2m

(C) r =0.5cm, ℓ =0.5m (D) r = 1 cm, ℓ = 1 m **EH0116**

101. One end of a conducting rod is maintained at temperature 50°C and at the other end, ice is melting at 0°C. The rate of melting of ice is doubled if:

(A) the temperature is made 200°C and the area of cross-section of the rod is doubled

(B) the temperature is made 100°C and length of rod is made four times

(C) area of cross-section of rod is halved and length is doubled

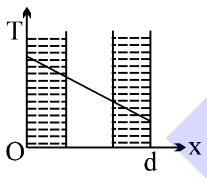
(D) the temperature is made 100°C and the area of cross-section of rod and length both are doubled.

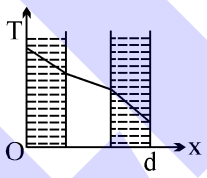
EH0121

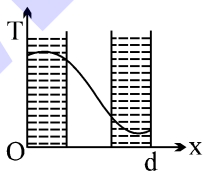
102. Water at 50°C is filled in a closed cylindrical vessel of height 10 cm and cross-sectional area 10 cm^2. The walls of the vessel are adiabatic but the flat parts are made of 1 mm thick aluminium (K = 200 J/m-s-s°C). Assume that the outside temperature is 20°C. The density of water is 1000 kg/m^3, and the specific heat capacity of water = 4200 J/kg–°C. Estimate the time taken for the temperature to fall by 1.0°C. (Assume heat current to be constant as drop in temperature is small)

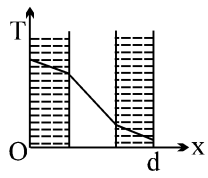
(A) 0.035 s (B) 0.040 s (C) 0.045 s (D) 0.050 s **EH0303**

103. The wall with a cavity consists of two layers of brick separated by a layer of air. All three layers have the same thickness and the thermal conductivity of the brick is much greater than that of air. The left layer is at a higher temperature than the right layer and steady state condition exists. Which of the following graphs predicts correctly the variation of temperature T with distance d inside the cavity?

(A) 

(B) 

(C) 

(D) 

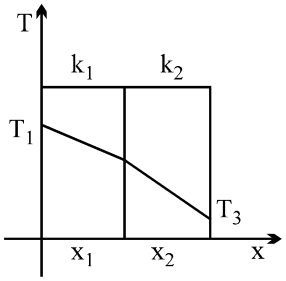
EH0077

104. A wall has two layer A and B each made of different material, both the layers have the same thickness. The thermal conductivity of the material A is twice that of B. Under thermal equilibrium the temperature difference across the wall B is 36°C. The temperature difference across the wall A is:

(A) 6°C (B) 12°C (C) 18°C (D) 72°C

EH0078

105. The temperature drop through each layer of a two layer furnace wall is shown in figure. Assume that the external temperature T_1 and T_3 are maintained constant and $T_1 > T_3$. If the thickness of the layers x_1 and x_2 are the same, which of the following statements are correct.



(A) $k_1 > k_2$

(B) $k_1 < k_2$

(C) $k_1 = k_2$ but heat flow through material (1) is larger then through (2)

(D) $k_1 = k_2$ but heat flow through material (1) is less than that through (2)

EH0081

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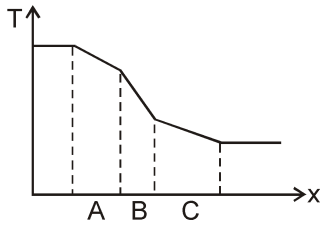
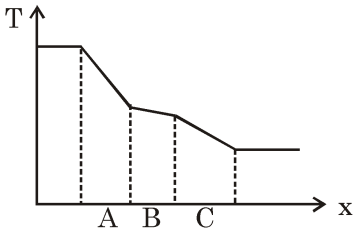
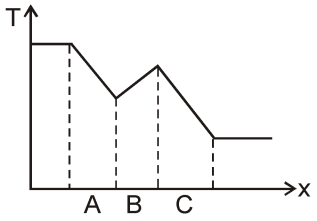
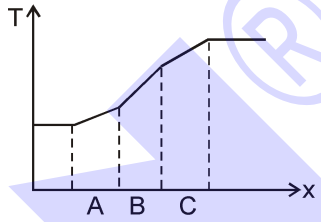
106. Graphs give the temperature along an x axis that extends directly through a wall consisting of three layers with different, nonzero and finite coefficient of thermal conductivities. The air temperature on both sides of the wall are different and uniform. Out of the situations as shown by the graphs, which is/are impossible ?

(A) 

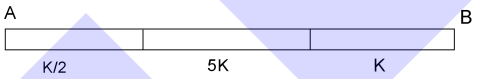
(B) 

(C) 

(D) 

EH0304

107. A composite rod made of three rods of equal length and cross-section as shown in the fig. The thermal conductivities of the materials of the rods are K/2, 5K and K respectively. The end A and end B are at constant temperatures. All heat entering the end A goes out of the end B, there being no loss of heat from the sides of the bar. The effective thermal conductivity of the bar is

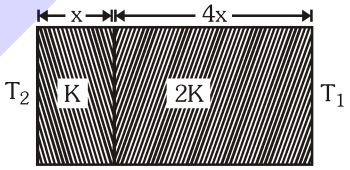


(A) 15K/16 (B) 6K/13 (C) 5K/16 (D) 2K/13.

EH0082

108. The temperature of the two outer surfaces of a composite slab, consisting of two materials having coefficients of thermal conductivity K and 2K and thickness x and 4x, respectively are T_2 and T_1($T_2 > T_1$). The rate of heat transfer through the slab, in a steady state is $\left(\frac{A(T_2 - T_1)K}{x}\right)$f, with f equals to-

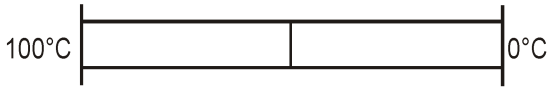
[AIEEE - 2004]



(A) 1 (B) 1/2 (C) 2/3 (D) 1/3

EH0084

109. The two rods shown in figure have identical geometrical dimensions. They are in contact with two heat baths at temperatures 100°C and 0°C. The temperature of the junction is 70°C. Find the temperature of the junction if the rods are interchanged.



(A) 70°C (B) 50°C (C) 30°C (D) can't determine

EH0305

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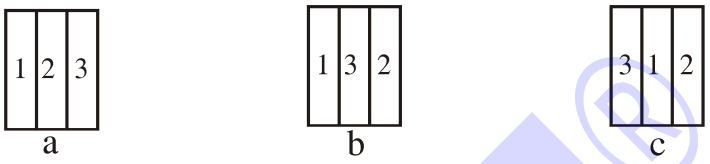
110. Two identical conducting rods are first connected independently to two vessels, one containing water at 100°C and the other containing ice at 0° C. In the second case, the rods are joined end to end and connected to the same vessels. Let q_1 and q_2 g/s be the rate of melting of ice in the two cases respectively. The ratio q_2/q_1 is: **[JEE' 2004 (Scr.)]**

(A) 1/2 (B) 2/1 (C) 4/1 (D) 1/4

EH0079

111. Figure shows three different arrangements of materials 1, 2 and 3 to form a wall. Thermal conductivities are $k_1 > k_2 > k_3$. The left side of the wall is 20°C higher than the right side. Temperature difference ΔT across the material 1 has following relation in three cases :



(A) $\Delta T_a > \Delta T_b > \Delta T_c$ (B) $\Delta T_a = \Delta T_b = \Delta T_c$ (C) $\Delta T_a = \Delta T_b > \Delta T_c$ (D) $\Delta T_a = \Delta T_b < \Delta T_c$

EH0083

112. An aluminium rod and a copper rod of equal length 1.0 m and cross-sectional area 1 cm^2 are welded together as shown in figure. One end is kept at a temperature of 20°C and the other at 60°C. Calculate the amount of heat taken out per second from the hot end. Thermal conductivity of aluminium = 200 W/m-°C and of copper = 390 W/m-°C.

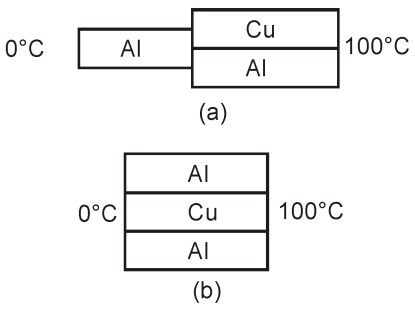
(A) 1.18 J (B) 2.36 J (C) 3.54 J (D) 4.72 J **EH0306**

113. A cylinder of radius R made of a material of thermal conductivity K_1 is surrounded by a cylindrical shell of inner radius R and outer radius 3R made of material of thermal conductivity K_2 . The ends of the combined system are maintained at two different temperatures. There is no loss of heat across the cylindrical surface and the system is in steady state. The effective thermal conductivity is

(A) $K_1 + K_2$ (B) $\frac{K_1K_2}{K_1+K_2}$ (C) $\frac{K_1+3K_2}{4}$ (D) $\frac{K_1+8K_2}{9}$

EH0307

114. The three rods shown in figure have identical geometrical dimensions. Find the ratio of rate of heat flow when the rods are joined as in [a] to that in [b]. Thermal conductivities of aluminium and copper are 200 W/m-°C and 400 W/m°C respectively.



(A) 3 : 16 (B) 16 : 3 (C) 3 : 8 (D) 8 : 3 **EH0308**

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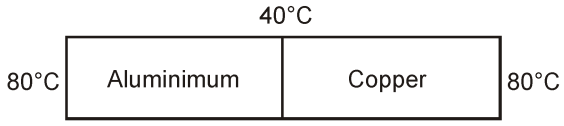
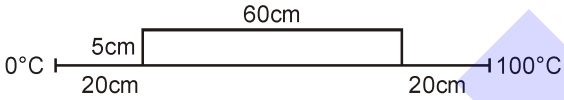
115. Figure shows an aluminium rod joined to a copper rod. Each of the rods has a length of 20 cm and area of cross-section 0.20 cm^2. The junction is maintained at a constant temperature 40°C and the two ends are maintained at 80°C. Calculate the amount of heat taken out from the cold junction in one minute after the steady state is reached. The conductivities are K_{Al} = 200 W/m-°C and K_{cu} = 400W/m-°C.



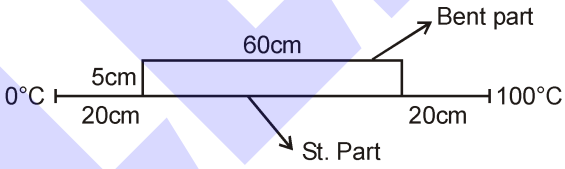
(A) 144 J (B) 192 J (C) 240 J (D) 288 J **EH0309**

116. Consider the situation shown in figure. The frame is made of the same material and has a uniform cross-sectional area everywhere.



(A) heat flowing through bent part is 60 J when 130 J is taken out from hot end.
(B) heat flowing through bent part is 30 J when 130 J is taken out from hot end.
(C) heat flowing through bent part is 60 J when 100 J is taken out from hot end.
(D) heat flowing through bent part is 30 J when 100 J is taken out from hot end. **EH0310**

117. Consider the situation shown in figure. The frame is made of the same material and has a uniform cross-sectional area everywhere. Suppose the bent part of the frame has a thermal conductivity of 780 $Jm^{-1}s^{-1}°C^{-1}$ whereas it is 390 J s^{-1} $m^{-1}°C^{-1}$for the straight part. Calculate the ratio of the rate of heat flow through the bent part to the rate of heat flow through the straight part.



(A) 12 : 7 (B) 7 : 12 (C) 13 : 6 (D) 6 : 13 **EH0311**

118. A closed cubical box is made of perfectly insulating material and the only way for heat to enter or leave the box is through two solid cylindrical metal plugs, each of cross sectional area 12 cm^2 and length 8 cm fixed in the opposite walls of the box. The outer surface of one plug is kept at a temperature of 100°C while the outer surface of the other plug is maintained at a temperature of 4°C. The thermal conductivity of the material of the plug is 2.0 $Wm^{-1}°C^{-1}$. A source of energy generating 13 W is enclosed inside the box. Find the equilibrium temperature of the inner surface of the box assuming that it is the same at all points on the inner surface.

(A) 249°C (B) 269°C (C) 289°C (D) 309°C **EH0312**

119. The ends of a metre stick are maintained at 100°C and 0°C. One end of a rod is maintained at 25°C. Where should its other end be touched on the metre stick so that there is no heat current in the rod in steady state?

(A) 15 cm from the cold end (B) 25 cm from the cold end
(C) 15 cm from the hot end (D) 25 cm from the hot end **EH0313**

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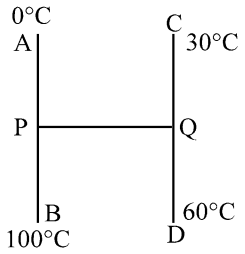
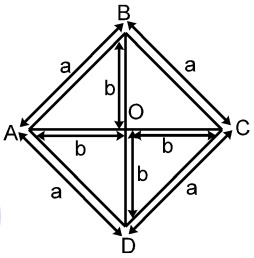
120. Three identical rods AB, CD and PQ are joined as shown. P and Q are mid points of AB and CD respectively. Ends A, B, C and D are maintained at 0°C, 100°C, 30°C and 60°C respectively. The direction of heat flow in PQ is



(A) from P to Q
(B) from Q to P
(C) heat does not flow in PQ
(D) data not sufficient

EH0080

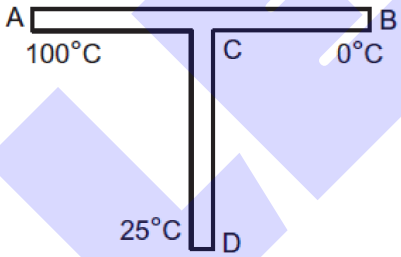
121. All the rods have same conductance 'K' and same area of cross section 'A'. If ends A and C are maintained at temperature $2T_0$ and T_0 respectively then which of the following is/are correct:



(A) Rate of heat flow through AB, AO and AD is same
(B) Rate of heat flow through BO and OD is not same
(C) Total Rate of heat flow from A to C is $\frac{3KAT_0}{2a}$
(D) Temperature at junctions B, O and D are same

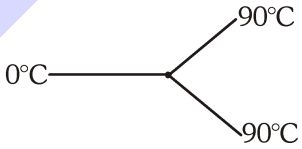
EH0314

122. A rod CD of thermal resistance 5.0 K W^{-1} is joined at the middle of an identical rod AB as shown in figure. The ends A, B and D are maintained at 100°C, 0°C and 25°C respectively. Find the heat current in CD.



(A) 4.0 W (B) 5.0 W (C) 6.0 W (D) 8.0 W **EH0315**

123. Three rods made of the same material and having the same cross-section have been joined as shown in the figure. Each rod have same length. The temperature of junction–



(A) 45°C (B) 60°C (C) 30°C (D) 20°C **EH0316**

124. A semicircular rod is joined at its end to a straight rod of the same material and the same cross-sectional area. The straight rod forms a diameter of the other rod. The junctions are maintained at different temperatures. Find the ratio of the heat transferred through a cross-sections of the semicircular rod to the heat transferred through a cross-section of the straight rod in a given time.

(A) π : 2 (B) 2 : π (C) 1 : π (D) π : 1 **EH0317**

125. A rod of negligible heat capacity has length 20 cm, area of cross-section 1.0 cm^2 and thermal conductivity 200 W/m-°C. The temperature of one end is maintained at 0°C and that of the other end is slowly and linearly varied from 0°C to 60°C in 10 minutes. Assuming no loss of heat through the sides, find the total heat transmitted through the rod in these 10 minutes.

(A) 600 J (B) 900 J (C) 1500 J (D) 1800 J **EH0318**

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126. A hollow metallic sphere of radius 20 cm surrounds a concentric metallic sphere of radius 5 cm. The space between the two spheres is filled with a nonmetallic material. The inner and outer spheres are maintained at 50°C and 10°C respectively and it is found that 100 J of heat passes from the inner sphere to the outer sphere per second. Find the thermal conductivity of the material between the spheres.
(A) 3.0 W/m–°C (B) 4.0 W/m–°C (C) 5.0 W/m–°C (D) 6.0 W/m–°C
EH0319

127. Steam is continuously passed through a 50 cm long rubber tube of inner and outer radii 1.0 cm and 1.2 cm maintaining its inner temperature at 120°C. The room temperature is 30°C. Calculate the rate of radial heat flow through the walls of the tube. Thermal conductivity of rubber = 0.15 J/m-s-°C.
(A) 183 J/s (B) 207 J/s (C) 233 J/s (D) 280 J/s **EH0320**

128. Two thin metallic spherical shells of radii r_1 and r_2 ($r_1 < r_2$) are placed with their centres coinciding. A material of thermal conductivity K is filled in the space between the shells. The inner shell is maintained at temperature θ_1 and the outer shell at temperature θ_2 ($\theta_1 < \theta_2$). Calculate the rate at which heat flows radially through the material.

(A) $\frac{4\pi K r_1 r_2 (\theta_2 - \theta_1)}{r_2 + r_1}$ (B) $\frac{2\pi K r_1 r_2 (\theta_2 - \theta_1)}{r_2 - r_1}$ (C) $\frac{4 K r_1 r_2 (\theta_2 - \theta_1)}{r_2 - r_1}$ (D) $\frac{4\pi K r_1 r_2 (\theta_2 - \theta_1)}{r_2 - r_1}$

EH0321

129. On a winter day when the atmospheric temperature drops to – 10°C, ice forms on the surface of a lake. Calculate the rate of increase of thickness of the ice when 10 cm of ice is already formed. Assume that the temperature of the entire water reaches 0°C before the ice starts forming. Density of water = 1000 kg/m^3, latent heat of fusion of ice = 3.36×10^5 J kg^{-1} and thermal conductivity of ice = 1.7 W/m–°C. Neglect the expansion of water on freezing.
(A) 4.5×10^{-7} m/s (B) 5.0×10^{-7} m/s
(C) 5.5×10^{-7} m/s (D) 6.0×10^{-7} m/s **EH0322**

130. On a winter day when the atmospheric temperature drops to – 10°C, ice forms on the surface of a lake. Calculate the approximate time taken in forming 10 cm of ice. Assume that the temperature of the entire water reaches 0°C before the ice starts forming. Density of water = 1000 kg/m^3, latent heat of fusion of ice = 3.36×10^5 J kg^{-1} and thermal conductivity of ice = 1.7 W/m–°C. Neglect the expansion of water on freezing.
(A) 27 hours (B) 13 hours (C) 22 hours (D) 17 hours **EH0323**

Radiation

131. The rate of emission of radiation of a black body at 273°C is E, then the rate of emission of radiation of this body at 0°C will be :-
(A) $\frac{E}{16}$ (B) $\frac{E}{4}$ (C) $\frac{E}{8}$ (D) 0 **EH0086**

132. Good absorber is good emitter is explained by–
(A) Stefan's law (B) Kirchoff's law
(C) Newton's law of cooling (D) Wein's displacement law **EH0324**

133. Two spheres of the same material have radii 1 m and 4 m and temperatures 4000 K and 2000 K respectively. The ratio of the energy radiated per second by the first sphere to that by the second is-
(A) 1 : 1 (B) 16 : 1 (C) 4 : 1 (D) 1 : 9
EH0089

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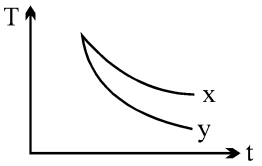
134. If emissivity of bodies X and Y are e_x and e_y and absorptive power are A_x and A_y then **[JEE' (Scr) 2003]**

(A) $e_y > e_x$; $A_y > A_x$
(B) $e_y < e_x$; $A_y < A_x$
(C) $e_y > e_x$; $A_y < A_x$
(D) $e_y = e_x$; $A_y = A_x$



EH0090

135. A metal ball of surface area 200 cm^2 and temperature 527°C is surrounded by a vessel at 27°C. If the emissivity of the metal is 0.4, then the rate of loss of heat from the ball is $(\sigma = 5.67 \times 10^{-8} J/m^2 - s - k^4)$

(A) 108 joules approx (B) 168 joules approx
(C) 182 joules approx (D) 192 joules approx **EH0325**

136. A blackbody of surface area 1 cm^2 is placed inside an enclosure. The enclosure has a constant temperature 27°C and the blackbody is maintained at 327° C by heating it electrically. What electric power is needed to maintain the temperature ? $\sigma = 6.0 \times 10^{-8}$ $Wm^{-2}K^{-4}$.

(A) 0.73 W (B) 0.85 W (C) 0.97 W (D) 1.21 W **EH0326**

137. An electric heater emits 1000 W of thermal radiation. The coil has a surface area of 0.020 m^2. Assuming that the coil radiates like a blackbody, find its temperature. $\sigma = 6.00 \times 10^{-8}$ $Wm^{-2}K^{-4}$.

(A) 955 °C (B) 955 K (C) 1025 °C (D) 1025 K **EH0327**

138. Two bodies A and B having equal surface area are maintained at temperatures 10°C and 20°C. The thermal radiation emitted in a given time by A and B are in the ratio

(A) 1 : 1.15 (B) 1 : 2 (C) 1 : 4 (D) 1 : 16 **EH0328**

139. Assume that the total surface area of a human body is 1.6 m^2 and that it radiates like an ideal radiator. Calculate the amount of energy radiated per second by the body if the body temperature is 37°C. Stefan constant σ is 6.0×10^{-8} W/m^2-K^4.

(A) 887 J (B) 967 J (C) 1033 J (D) 1107 J **EH0329**

140. A cubical block of mass 1.0 kg and edge 5.0 cm is heated to 227°C. It is kept in an evacuated chamber maintained at 27°C. Assuming that the block emits radiation like a blackbody, find the rate at which the temperature of the block will decrease. Specific heat capacity of the material of the block is 400 J/kg-K.

(A) 0.09°C/s (B) 0.12°C/s (C) 0.15°C/s (D) 0.18°C/s **EH0330**

141. A spherical ball of surface area 20 cm^2 absorbs any radiation that falls on it. It is suspended in a closed box maintained a 57°C. Find the amount of radiation falling on the ball per second. Stefan constant = 6.0×10^{-8} W/m^2–K^4.

(A) 1.4 J (B) 1.8 J (C) 2.2 J (D) 2.4 J **EH0331**

142. A black metal foil is warmed by radiation from a small sphere at temperature 'T' and at a distance 'd'. It is found that the power received by the foil is P. If both the temperature and distance are doubled, the power received by the foil will be :

(A) 16 P (B) 4 P (C) 2 P (D) P **EH0085**

143. A sphere and cube of same material and same volume are heated upto the same temperature and allowed to cool in the same surroundings. The ratio of the amounts of radiations emitted will be

(A) 1 : 1 (B) $\frac{4\pi}{3} : 1$ (C) $\left(\frac{\pi}{6}\right)^{1/3} : 1$ (D) $\frac{1}{2}\left(\frac{4\pi}{3}\right)^{2/3} : 1$

EH0332

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144. Spheres P and Q are uniformly constructed from the same material which is a good conductor of heat and the radius of Q is thrice the radius of P. The rate of fall of temperature of P is x times that of Q when both are at the same surface temperature. The value of x is :

(A) 1/4 (B) 1/3 (C) 3 (D) 4 **EH0088**

145. If the temperature of the sun were to increase from T to 2T and its radius from R to 2R, then the ratio of the radiant energy received on earth to what it was previously, will be- **[AIEEE - 2004]**

(A) 4 (B) 16 (C) 32 (D) 64

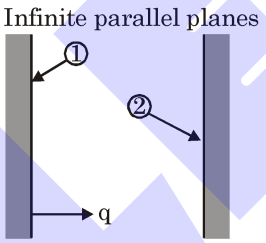
EH0094

146. A long solid cylinder is radiating power. It is remoulded into a number of smaller cylinders, each of which has the same length as original cylinder. Each small cylinder has the same temperature as the original cylinder. The total radiant power emitted by the pieces is twice that emitted by the original cylinder. How many smaller cylinders are there ? Neglect the energy emitted by the flat faces of cylinder.

(A) 3 (B) 4 (C) 5 (D) 6

EH0115

147. In the figure shown below, there are two large planes maintained at temperature of 27° C and 127 ° C respectively. The surfaces are completely black. Taking Stefan's constant to be $\frac{17}{3}\times10^{-8}\,W/m^2K^4$, what is the rate of heat transfer per unit area between the plates?

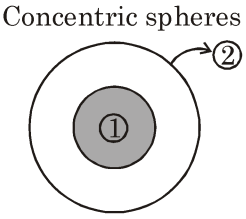


(A) 992 W/m^2 (B) 14.71 W/m^2 (C) 446 W/m^2 (D) 29.5 W/m^2 **EH0333**

148. In the previous case, if each of the planes is a grey body with emissivity 0.5, by what factor is the rate of heat transfer per unit area between the plates changed?

(A) ½ (B) 1/3 (C) ¼ (D) 1/5 **EH0334**

149. We have two concentric black spheres of radii R_1 and R_2 as shown. If the spheres are at temperatures T_1 and T_2 respectively, what is the rate of heat flow between them?



(A) $4\pi\sigma R_1^2T_1^4 - 4\pi\sigma R_2^2T_2^4$ (B) $4\pi\sigma R_1^2\left(T_1^4 - T_2^4\right)$

(C) $4\pi\sigma R_2^2\left(T_1^4 - T_2^4\right)$ (D) $4\pi\sigma R_1R_2\left(T_1^4 - T_2^4\right)$ **EH0335**

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150. If these spheres are not concentric, the rate of heat transfer

(A) Increases (B) Decreases

(C) Remains the same (D) Cannot be predicted **EH0336**

151. Three very large plates of same area are kept parallel and close to each other. They are considered as ideal black surfaces and have very high thermal conductivity. The first and third plates are maintained at temperatures 2T and 3T respectively. The temperature of the middle (i.e. second) plate under steady state condition is **[JEE 2012]**

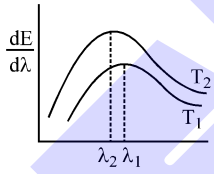
(A) $\left(\frac{65}{2}\right)^{1/4} T$ (B) $\left(\frac{97}{4}\right)^{1/4} T$ (C) $\left(\frac{97}{2}\right)^{1/4} T$ (D) $(97)^{1/4} T$

EH0147

152. Total radiations incident on body = 400 J, 20% radiation reflected and 120 J absorbs. Then find out % of transmittive power

(A) 10 (B) 15 (C) 25 (D) 50 **EH0337**

153. The spectral emissive power E_λ for a body at temperature T_1 is plotted against the wavelength and area under the curve is found to be A. At a different temperature T_2 the area is found to be 9A. Then λ_1/λ_2 =



(A) 3 (B) 1/3 (C) $1/\sqrt{3}$ (D) $\sqrt{3}$

EH0118

154. The spectrum of a blackbody peaks at a wavelengtgh inversely proportional to its temperature. This is knw as Wein's law and is used to estimate stellar temperatures. The sun can be approximated as a blackbody with its peak wavelength in the visible portion of the spectrum and a surface temperature of 6000 K. Given this data, estimate the peak wavelength of a human being, assuming it to be a black body.

(A) 10 nm (B) 10 μm (C) 10 mm (D) 10 m **EH0338**

155. If the surface temperature of a star is 6000 K, in which colour would it appear in a colour photograph ?

(A) Red (B) Cyan (C) Yellow (D) Blue **EH0339**

156. A black body is at a temperature of 2880 K. The energy of radiation emitted by this object with wavelength between 499 nm and 500 nm is U_1, between 999 nm and 1000 nm is U_2 and between 1499 nm and 1500 nm is U_3. The Wien constant b = 2.88×10^6 nm K. Then

(A) $U_1 = 0$ (b) $U_3 = 0$ (C) $U_1 > U_2$ (D) $U_2 > U_1$

EH0122

157. The power radiated by a black body is P and it radiates maximum energy around the wavelength λ_0. If the temperature of the black body is now changed so that it radiates maximum energy around wavelength $3/4\lambda_0$, the power radiated by it will increase by a factor of :-

(A) 4/3 (B) 16/9 (C) 64/27 (D) 256/81

EH0087

158. Three discs A, B, and C having radii 2 m, 4 m and 6 m respectively are coated with carbon black on their outer surfaces. The wavelengths corresponding to maximum intensity are 300 nm, 400 nm and 500 nm respectively. The power radiated by them are Q_A, Q_B and Q_C respectively.

[JEE' 2004 (Scr.)]

(A) Q_A is maximum (B) Q_B is maximum

(C) Q_C is maximum (D) $Q_A = Q_B = Q_C$ **EH0091**

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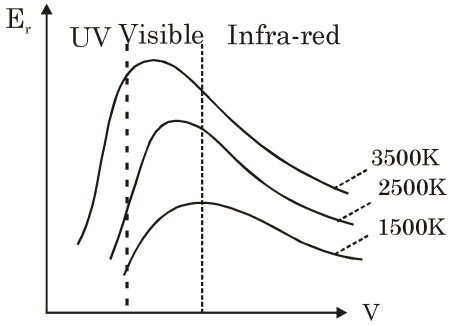
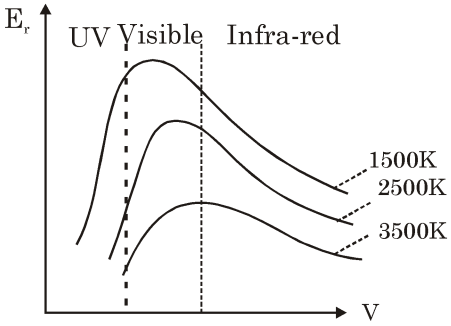
159. Which of the following graphs shows the correct variation in intensity of heat radiations by black body and frequency at a fixed temperature?

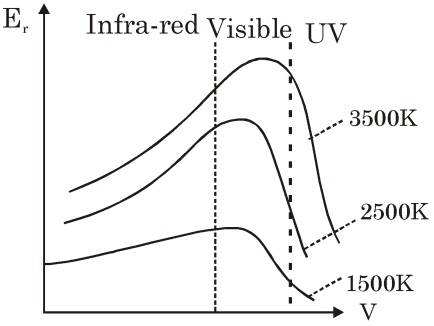
(A)



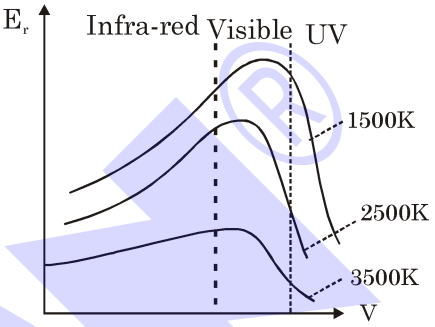
(B)



(C)



(D)



EH0340

160. A black body calorimeter filled with hot water cools from 60°C to 50°C in 4 min and 40°C to 30°C in 8 min. The approximate temperature of surrounding is :

(A) 10°C (B) 15°C (C) 20°C (D) 25°C **EH0092**

161. The temperature of a body falls from 40°C to 36°C in 5 minutes when placed in a surrounding of constant temperature 16°C. Find the time taken for the temperature of the body to become 32°C.

(A) 5 min (B) 5.5 min (C) 6.1 min (D) 7 min **EH0341**

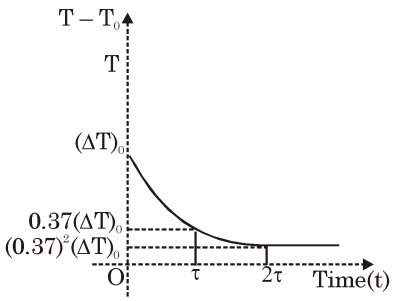
162. A calorimeter contains 50 g of water at 50°C. The temperature falls to 45°C in 10 minutes. When the calorimeter contains 100 g of water at 50°C, it takes 18 minutes for the temperature to becomes 45°C. Find the water equivalent of the calorimeter.

(A) 12.5 g (B) 15.0 g (C) 17.5 g (D) 20.0 g **EH0342**

163. A body takes 5 minute to cool from 65°C to 60°C and 7 minutes to cool from 60°C to 55°C. Find out the room temperature and the time it will take to cool from 55°C to 50°C.

(A) 40°C, 10 min (B) 35°C, 11.67 min
(C) 45°C, 11.67 min (D) 40°C, 15 min **EH0343**

164. A calorimeter of negligible heat capacity contains 100 cc of water at 40° C. The water cools to 35°C in 5 min. The water is now replaced by k-oil A equal volume at 40°C, Find the time taken for the temp to become 35°C (Given densities of water and K-oil are respectively 1000 and 800 kg.m^{-3}; and their specific heats are respectively: 4200 and 2100 J/kg-°C)



(A) 1 min (B) 2 min (C) 3 min (D) 4 min **EH0344**

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165. A radiator whose temperature is T° C, is used to heat the room in the cold weather. The radiator is able to maintain a room temperature of 30° C when outside temperature is –10°C and 15°C when outside temperature is –30°C. Determine the temperature of the radiator [Assume Newton's law of cooling to be valid]

(A) 85° C (B) 15 °C (C) 98.6 °C (D) 150°C **EH0345**

166. A solid copper sphere of dimater 10mm, is cooled to a temperature of 150K and is then placed in an enclosure at 290 K. Assuming that all interchange of heat is by radiation, calculate the initial rate of rise of temperature of the sphere. The sphere may be treated as a black body $\rho_{copper} = 8.93 \times 10^3$ kg/m^3, s = 3.7×10^2 $JKg^{-2}K^{-1}$; $\sigma = 5.7 \times 10^{-8}$ $Wm^{-2}K^{-4}$

(A) 0.68 K/s (B) 0.068 K/s (C) 0.34 K/s (D) 0.034 K/s **EH0346**

167. A spherical tungsten piece of radius 1.0 cm is suspended in an evacuated chamber maintained at 300 K. The piece is maintained at 1000 K by heating it electrically. Find the rate at which the electrical energy must be supplied. The emissivity of tungsten is 0.30 and the Stefan constant σ is 6.0×10^{-8} $W/m^2–K^4$.

(A) 18 W (B) 22 W (C) 26 W (D) 30 W **EH0347**

Multiple Correct Questions

Elasticity

168. A wire is suspended from the ceiling and stretched under the action of a weight *F* suspended from its other end. The force exerted by the ceiling on it is equal and opposite to the weight.

(A) Tensile stress at any cross section *A* of the wire is *F/A*.

(B) Tensile stress at any cross section is zero.

(C) Tensile stress at any cross section *A* of the wire is *2F/A*.

(D) Tension at any cross section A of the wire is *F*. **EH0095**

169. A body of mass M is attached to the lower end of a metal wire, whose upper end is fixed. The elongation of the wire is ℓ.

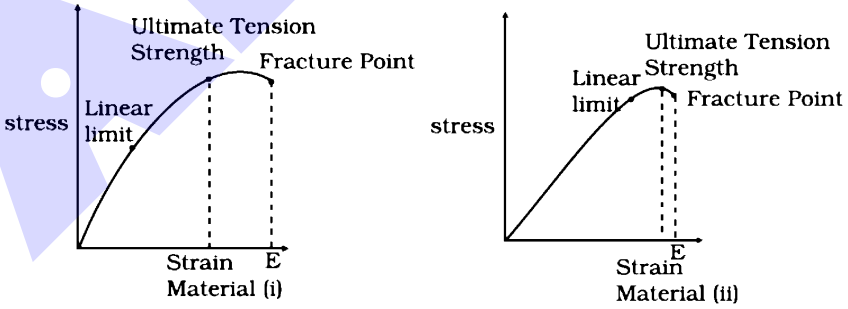
(A) Loss in gravitational potential energy of M is Mgℓ

(B) The elastic potential energy stored in the wire is Mgℓ

(C) The elastic potential energy stored in the wire is 1/2 Mgℓ

(D) Heat produced is 1/2 Mgℓ. **EH0097**

170. The stress-strain graphs for two materials are shown in figure (assume same scale).



(A) Material (ii) is more elastic than material (i) and hence material (ii) is more brittle.

(B) Material (i) and (ii) have the same elasticity and the same brittleness.

(C) Material (ii) is elastic over a larger region of strain as compared to (i).

(D) Material (ii) is more brittle than material (i). **EH0098**

171. The torque required to produce a unit twist in a solid bar of length L and radius r is

(A) directly proportional to r^2 (B) directly proportional to r^4

(C) inversely proportional to l (D) inversely proportional to r^2 **EH0348**

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Thermal expansion

172. When the temperature of a copper coin is raised by 80°C, its diameter increases by 0.2%.
(A) Percentage rise in the area of a face is 0.4 %
(B) Percentage rise in the thickness is 0.4 %
(C) Percentage rise in the volume is 0.6 %
(D) Coefficient of linear expansion of copper is 0.25×10^{-4} C° $^{-1}$. **EH0103**

173. A meter scale is made up of steel and measures correct length at 16°C. What will be the percentage error respectively if this scale is used (a) on a summer day when the temperature is 46°C and (b) on a winter day when the temperature is 6°C? Cofficient of linear expansion of steel = 11×10^{-6}/°C.
(A) Percentage error in the reading of scale at 46°C is 0.033 %
(B) Percentage error in the reading of scale at 46°C is – 0.033 %
(C) Percentage error in the reading of scale at 6°C is 0.011 %
(D) Percentage error in the reading of scale at 6°C is – 0.011 % **EH0349**

Calorimetry

174. Two identical beakers are filled with water to the same level at 4°C. If one say A is heated while the other B is cooled, then:–
(A) water level in A will rise (B) water level in B will rise
(C) water level in A will fall (D) water level in B will fall **EH0350**

175. Mark the **CORRECT** options:
(A) A system *X* is in thermal equilibrium with *Y* but not with *Z*. System *Y* and *Z* may be in thermal equilibrium with each other.
(B) A system *X* is in thermal equilibrium with *Y* but not with *Z*. Systems *Y* and *Z* are not in thermal equilibrium with each other.
(C) A system *X* is neither in thermal equilibrium with *Y* nor with *Z*. The systems *Y* and *Z* must be in thermal equilibrium with each other.
(D) A system *X* is neither in thermal equilibrium with *Y* nor with *Z*. The system *Y* and *Z* may be in thermal equilibrium with each other.
EH0101

176. 50 gm ice at – 10°C is mixed with 20 gm steam at 100°C. When the mixture finally reaches its steady state inside a calorimeter of water equivalent 1.5 gm then : [Assume calorimeter was initially at 0°C, Take latent heat of vaporization of water = 540 cal/gm, Latent heat of fusion of water = 80 cal/gm, specific heat capacity of water = 1 cal/gm-°C, specific heat capacity of ice = 0.5 cal/gm°C]
(A) Mass of water remaining is : 67.4 gm
(B) Mass of steam remaining is : 2.6 gm
(C) Mass of water remaining is : 67.87 gm
(D) Mass of steam remaining is : 2.13 gm **EH0102**

177. Four 2 cm × 2 cm × 2 cm cubes of ice are taken out from a refrigerator and are put in 200 ml of a drink at 10°C. Assume that no heat is lost to the outside of the drink and that the container has negligible heat capacity. Density of ice = 900 kg/m^3, density of the drink = 1000 kg/m^3, specific heat capacity of the drink = 4200 J/kg-K, latent heat of fusion of ice = 3.4×10^5 J/kg.
(A) Final temperature of the drink is 0°C
(B) Final temperature of the drink is 1.1 °C
(C) Amount of ice melted is 28.8 g
(D) Amount of ice melted is 25 g **EH0351**

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Conduction

178. A bar of copper of length 75 cm and a bar of steel of length 125 cm are joined together end to end. Both are of circular cross section with diameter 2 cm. The free ends of the copper and the steel bars are maintained at 100°C and 0°C respectively. The curved surfaces of the bars are thermally insulated. Thermal conductivity of copper is 386 $Js^{-1}\ m^{-1}°C^{-1}$ and that of steel is 46 $J\ s^{-1}m^{-1}°C^{-1}$.

(A) Temperature of the junction is 7 °C

(B) Temperature of the junction is 93 °C

(C) Amount of heat transmitted across the junction per unit time is 1.07 Js^{-1}

(D) Amount of heat transmitted across the junction per unit time is 2.14 Js^{-1} **EH0352**

Radiation

179. A piece of charcoal and a piece of shining steel of the same area are kept for long time in an open lawn in bright sun.

(A) The steel will absorb more heat than the charcoal.

(B) The temperatuer of the steel will be higher than that of the charcoal.

(C) If the both are picked up by bare hands, the steel will be felt hotter than the charcoal

(D) If the two are picked up from the lawn and kept in a cold chamber, the charcoal will lose heat at a faster rate than the steel **EH0353**

180. Two metallic sphere A and B are made of same material and have got identical surface finish. The mass of sphere A is four times that of B. Both the spheres are heated to the same temperature and placed in a room having lower temperature but thermally insulated from each other.

(A) The ratio of heat loss of A to that of B is $2^{4/3}$.

(B) The ratio of heat loss of A to that of B is $2^{2/3}$.

(C) The ratio of the initial rate of cooling of A to that of B is $2^{-2/3}$.

(D) The ratio of the initial rate of cooling of A to that of B is $2^{-4/3}$. **EH0104**

181. Two bodies A and B have thermal emissivities of 0.01 and 0.81 respectively. The outer surface areas of the two bodies are the same. The two bodies radiate energy at the same rate. The wavelength λ_B, corresponding to the maximum spectral radiancy in the radiation from B, is shifted from the wavelength corresponding to the maximum spectral radiancy in the radiation from A by 1.00 μm. If the temperature of A is 5802 K,

(A) the temperature of B is 1934 K (B) λ_B =1.5 μm

(C) the temperature of B is 11604 K (D) the temperature of B is 2901 K

EH0105

182. A heated body emits radiation which has maximum intensity near the frequency v0. The emissivity of the material is 0.5. If the absolute temperature of the body is doubled

(A) the maximum intensity of radition will be near the frequency 2v0

(B) the maximum intensity of radiation will be near the frequency v0/2

(C) the total energy emitted will increase by a factor of 16

(D) the total energy emitted will increase by a factor of 8 **EH0354**

183. A solid sphere and a hollow sphere of the same material and of equal radii are heated to the same temperature.

(A) Both will emit equal amount of radiation per unit time in the biginning

(B) Both will absorb equal amount of radiation from the surrounding in the beginning

(C) The initial rate of colling (dT/dt) will be the same for the two spheres

(D) The two spheres will have equal temperatures at any instant. **EH0355**

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184. A metal cylinder of mass 0.5 kg is heated electrically by a 12 W heater in a room at 15°C. The cylinder temperature rises to 25°C in 5 min and finally becomes constant at 45°C. Asuming that the rate of heat loss is proportional to the excess temperature over the surroundings.

(A) The rate of loss of heat of the cylinder to surrounding ar 20°C is 2 W

(B) The rate of loss of heat of the cylinder to surrounding at 45°C is 2 W

(C) Specific heat capacity of metal is $\frac{240}{\ln(3/2)}$ J / kg°C

(D) None fof these **EH0356**

185. When we consider convection with radiation in Newton's law of cooling, choose correct statements about rate of heat loss.

(A) directly proportional to emissivity

(B) directly proportional to Stefan's constant

(C) directly proportional to surface area

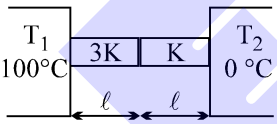
(D) directly proportional to temperature difference of body and room. **EH0357**

COMPREHENSION TYPE QUESTIONS

Conduction

Paragraph for Question No. 186 to 188

Two rods A and B of same cross-sectional are A and length ℓ connected in series between a source (T_1 = 100°C) and a sink (T_2 = 0°C) as shown in figure. The rod is laterally insulated.



186. The ratio of the thermal resistance of the rods is

(A) $\frac{R_A}{R_B} = \frac{1}{3}$ (B) $\frac{R_A}{R_B} = 3$ (C) $\frac{R_A}{R_B} = \frac{3}{4}$ (D) $\frac{4}{3}$ **EH0106**

187. If T_A and T_B are the temperature drops across the rod A and B, then

(A) $\frac{T_A}{T_B} = \frac{3}{1}$ (B) $\frac{T_A}{T_B} = \frac{1}{3}$ (C) $\frac{T_A}{T_B} = \frac{3}{4}$ (D) $\frac{T_A}{T_B} = \frac{4}{3}$ **EH0106**

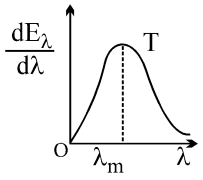
188. If G_A and G_B are the temperature gradients across the rod A and B, then

(A) $\frac{G_A}{G_B} = \frac{3}{1}$ (B) $\frac{G_A}{G_B} = \frac{1}{3}$ (C) $\frac{G_A}{G_B} = \frac{3}{4}$ (D) $\frac{G_A}{G_B} = \frac{4}{3}$ **EH0106**

Paragraph for Question No. 189 and 190

Radiation

The figure shows a radiant energy spectrum graph for a black body at a temperature T.



189. Choose the **CORRECT** statement(s)

(A) The radiant energy is not equally distributed among all the possible wavelengths

(B) For a particular wavelength the spectral intensity is maximum

(C) The area under the curve is equal to the total rate at which heat is radiated by the body at that temperature.

(D) None of these **EH0124**

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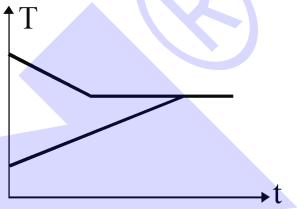
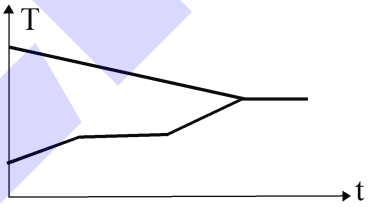
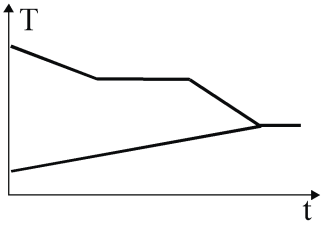
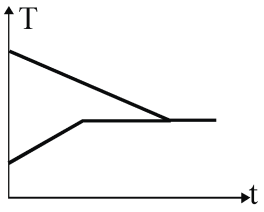
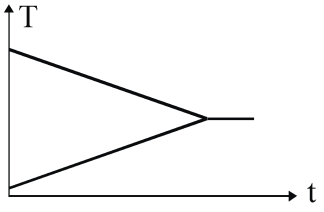
190. If the temperature of the body is raised to a higher temperature T', then choose the correct statement(s)
(A) The intensity of radiation for every wavelength increases
(B) The maximum intensity occurs at a shorter wavelength
(C) The area under the graph increases
(D) The area under the graph is proportional to the fourth power of temperature

EH0124

Calorimetry

191. A sample 'A' of liquid water and a sample B of ice of equal mass are kept in 2 nearby containers so that they can exchange heat with each other but are thermally insulated from the surroundings. The graphs in column-II show the sketch of temperature T of samples versus time t. Match with appropriate description in column-I.

Column I		**Column II**
(A) Equilibrium temperature is above melting point of ice.	(P)	
(B) At least some of water freezes.	(Q)	
(C) At least some of ice melts.	(R)	
(D) Equilibrium temperature is below freezing point of water	(S)	
	(T)	

EH0126

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EXERCISE (O-2)

Elasticity

1. A cylindrical wire of radius 1 mm, length 1 m, Young's modulus = 2×10^{11} N/m^2, poisson's ratio $\mu = \pi/10$ is stretched by a force of 100 N. Its radius will become.

(A) 0.99998 mm (B) 0.99999 mm (C) 0.99997 mm (D) 0.99995 mm

EH0107

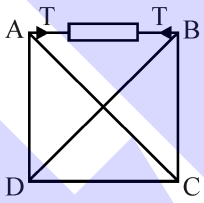
2. A uniform plank of Young's modulus Y is moved over a smooth horizontal surface by a constant horizontal force F. The area of cross section of the plank is A. The compressive strain on the plank in the direction of the force is

(A) F / AY (B) 2F / AY (C) $\frac{1}{2}$(F / AY) (D) 3F / AY **EH0358**

3. A metallic ring of radius 2cm and cross sectional area 4cm^2 is fitted into a wooden circular disc of radius 4cm. If the Young's modulus of the material of the ring is 2×10^{11}N/m^2, the force with which the metal ring expands is:

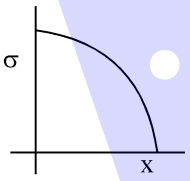
(A) 2×10^7 N (B) 8×10^7 N (C) 4×10^7 N (D) 6×10^7 N **EH0359**

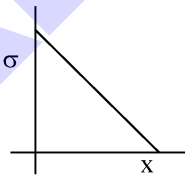
4. A rigid square frame consists of four side bars AB, BC, CD and DA and two diagonal bars AC and BD which are only touching each other freely at centre. They are hinged at the joints. By some arrangement AB is kept under tension so that AB has tension T at its ends as shown. Assume symmetry.

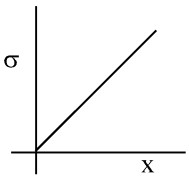


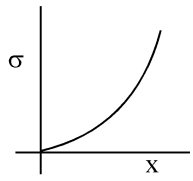
(A) AD and BC are also under tension. (B) AD and BC are under compression.
(C) AC and BD are under tension. (D) CD is under compression. **EH0360**

5. A uniform rod rotating in gravity free region with certain constant angular velocity. The variation of tensile stress with distance x from axis of rotation is best represented by which of the following graphs.

(A) 

(B) 

(C) 

(D) 

EH0048

6. A highly rigid cubical block A of small mass M and side L is fixed rigidly on another cubical block B of same dimensions and of low modulus of rigidity η, such that the lower face of A completely covers the upper face of B. The lower face of B is rigidly held on a horizontal surface. A small force F is applied perpendicular to one of the side faces of A at the top. The side of B turns by an angle ϕ. Now the whole arrangement is turned upside down so that lower face of A is rigidly held to the horizontal surface and we apply a small force F perpendicular to the side face of B at the top. The Side of B now turns by an angle

(A) ϕ (B) $\phi/2$ (C) $\phi/3$ (D) zero **EH0361**

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Thermal expansion

7. A rod has variable co-efficient of linear expansion $\alpha = \frac{x}{5000}$. If length of the rod is 1m. Determine increase in length of the rod on increasing temperature of the rod by 100°C.

(A) 1 cm (B) 0.5 cm (C) 1.5 cm (D) 0.75 cm **EH0362**

8. A thin circular metal disc of radius 500.0 mm is set rotating about a central axis normal to its plane. Upon raising its temperature gradually, the radius increases to 507.5 mm. The percentage change in the rotational kinetic energy will be

(A) 1.5% (B) –1.5% (C) 3% (D) –3% **EH0363**

9. An iron ring measuring 15.00 cm in diameter is to be shrunk on a pulley which is 15.05 cm in diameter. All measurments refer to the room temperature 20°C. The minimum temperature should the ring be heated to make the job possible and the strain developed in the ring when it comes to the room temperature are (Coefficient of linear expansion of iron = 12×10^{-6}/°C)

(A) 278°C (B) 298°C

(C) strain = 3.33×10^{-3} (D) strain = 6.67×10^{-3} **EH0364**

10. The volume of the bulb of a mercury thermometer at 0°C is V_0 and cross section of the capillary is A_0. The coefficient of linear expansion of glass is α_g per °C and the cubical expansion of mercury γ_m per °C. If the mercury just fills the bulb at 0°C, what is the length of mercury column in capillary at T°C.

(A) $\frac{V_0T(\gamma_m+3\alpha_g)}{A_0(1+2\alpha_gT)}$ (B) $\frac{V_0T(\gamma_m-3\alpha_g)}{A_0(1+2\alpha_gT)}$ (C) $\frac{V_0T(\gamma_m+2\alpha_g)}{A_0(1+3\alpha_gT)}$ (D) $\frac{V_0T(\gamma_m-2\alpha_g)}{A_0(1+3\alpha_gT)}$

EH0112

11. A bimetallic strip is formed out of two identical strips one of copper and the other of brass. The coefficient of linear expansion of the two metals are α_C and α_B. On heating, the temperature of the strip goes up by ΔT and the strip bends to form an arc of radius of curvature R. Then R is

(A) proportional at ΔT (B) inversely proportional to ΔT

(C) proportional to $|\alpha_B-\alpha_C|$ (D) inversely proportional to $|\alpha_B-\alpha_C|$

EH0123

Calorimetry

12. A thermally isolated vessel contains 100 g of water at 0°C. When air above the water is pumped out, some of the water freezes and some evaporates at 0°C itself. Calculate the mass of the ice formed if no water is left in the vessel. Latent heat of vaporization of water at 0°C = 2.10×10^6 J kg^{-1} and latent heat of fusion of ice = 3.36×10^5 J kg^{-1}.

(A) 40 g (B) 78 g (C) 86 g (D) 94 g **EH0365**

13. Find the amount of heat supplied to decrease the volume of an ice water mixture by 1 cm^3 without any change in temperature. ($\rho_{ice} = 0.9\ \rho_{water}$, L_{ice} = 80 cal/gm).

(A) 360 cal (B) 500 cal (C) 720 cal (D) none of these

EH0063

14. A thermally insulated vessel contains some water at 0^0C. The vessel is connected to a vacuum pump to pump out water vapour. This results in some water getting frozen. It is given Latent heat of vaporization of water at 0°C = 21×10^5 J/kg and latent heat of freezing of water = 3.36×10^5 J/kg. The maximum percentage amount of water that will be solidified in this manner will be :-

(A) 86.2% (B) 33.6% (C) 21% (D) 24.36%

EH0108

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15. A liquid is given some heat.

Statement A : Some liquid evaporates. **Statement B** : The liquid starts boiling.

(A) A implies B and B implies A (B) B implies A but, A does not imply B

(C) A implies B but B does not imply A (D) Neither A implies B nor B implies A

EH0114

16. When a small ice crystal is placed into super cooled water, it begins to freeze instantaneously. What amount of ice is formed from 1kg of water super cooled to –8° C .

(A) 40 g (B) 60 g (C) 80 g (D) 100 g **EH0366**

17. An experiment is perfomed to measure the specific heat of copper. A lump of copper is heated in an oven, then dropped into a beaker of water. To calculate the specific heat of copper, the experimenter must know or measure the value of all of the quantities below EXCEPT the

(A) heat capacity of water and beaker

(B) original temperature of the copper and the water

(C) final (equilibrium) temperature of the copper and the water

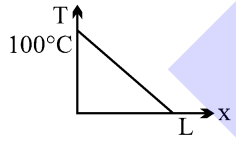
(D) time taken to achieve equilibrium, after the copper is dropped into the water. **EH0120**

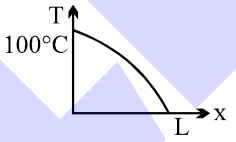
Conduction

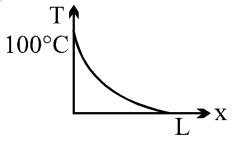
18. Two thin walled spheres of different materials, one with double the radius and one-fourth wall thickness of the other, are filled with ice. If the time taken for complete melting of ice in the sphere of larger radius is 25 minutes and that for smaller one is 16 minutes, the ratio of thermal conductivities of the materials of larger sphere to the smaller sphere is :

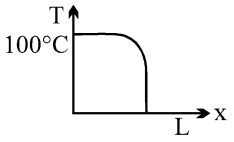
(A) 4 : 5 (B) 25 : 1 (C) 1 : 25 (D) 8 : 25 **EH0367**

19. A rod of length L and uniform cross-sectional area has varying thermal conductivity which changes linearly from 2K at end A to K at the other end B. The ends A and B of the rod are maintained at constant temperature 100°C and 0°C, respectively. At steady state, the graph of temperature : T = T(x) where x = distance from end A will be

(A) 

(B) 

(C) 

(D) 

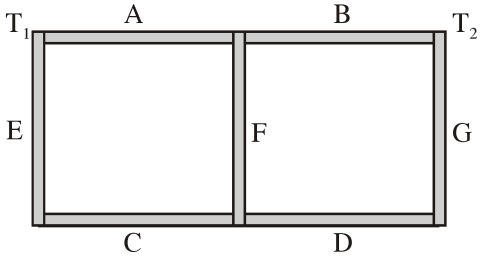
EH0117

20. Seven rods, A, B, C, D, E, F and G are joined as shown in figure. All the rods have equal cross-sectional area A and length l. The thermal conductivities of the rods are $K_A = K_C = K_O$, $K_B = K_D = 2K_O$, $K_E = 3K_O$, $K_F = 4K_O$ and $K_G = 5K_O$. The rod E is kept at a constant temperature T_1 and the rod G is kept at a constant temperature T_2 $(T_2 > T_1)$

(A) The rod F has non uniform temperature.

(B) The rod F has uniform temperature of $\frac{(T_1 + 2T_2)}{3}$.

(C) The rate of heat flow from the source which is maintains the temperature T_2 is $\frac{4K_0A(T_2 - T_1)}{3\ell}$.



(D) The rate of heat flow from the source which is maintains the temperature T_2 is $\frac{2K_0A(T_2 - T_1)}{\ell}$.

EH0368

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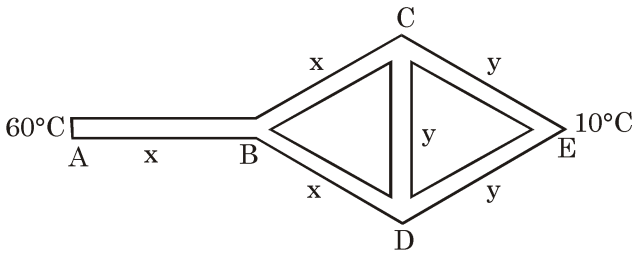
21. Three rods of material x and three of material y are connected as shown in figure. All the rods are identical in length and cross sectional area. If the end A is maintained at 60°C and the junction E at 10°C, calculate the temperature of the junction B. The thermal conductivity of x is 800 W $m^{-1}°C^{-1}$ and that of y is 400 W $m^{-1}°C^{-1}$.



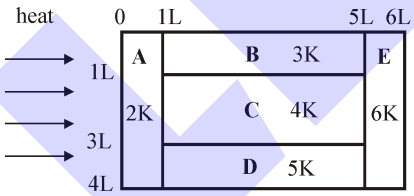
(A) 30°C (B) 35°C (C) 40°C (D) 45°C **EH0369**

22. A metal rod of cross-sectional area 1.0 cm^2 is being heated at one end. At one time, the temperature gradient is 5.0°C/cm at cross-section A and is 2.5°C/cm at cross-section B. Calculate the rate at which the temperature is increasing in the part AB of the rod. The heat capacity of the part AB = 0.40 J°C, thermal conductivity of the material of the rod = 200 W/m-°C. Neglect any loss of heat to the atmosphere.

(A) 10 °C/s (B) 12.5°C/s (C) 15°C/s (D) 17°C/s **EH0370**

23. A composite block is made of slabs A, B, C, D and E of different thermal conductivities (given in terms of a constant K) and sizes (given in terms of length, L) as shown in the figure. All slabs are of same width. Heat 'Q' flows only from left to right through the blocks. Then in steady state.

[JEE 2011]



(A) heat flow through A and E slabs are same.
(B) heat flow through slab E is maximum.
(C) temperature difference across slab E is smallest.
(D) heat flow through C= heat flow through B + Heat flow through D. **EH0145**

24. In a 20m deep lake, the bottom is at a constant temperature of 4°C. The air temperature is constant at –10°C. The thermal conductivity of ice is 4 times that water. Neglecting the expansion of water on freezing, the maximum thickness of ice will be

(A) $\frac{20}{11}$ m (B) $\frac{200}{11}$ m (C) 20 m (D) 10 m **EH0371**

Radiation

25. 'Gulab Jamuns' (assumed to be spherical) are to be heated in an oven. They are available in two sizes, one twice bigger (in radius) than the other. Pizzas (assumed to be discs) are also to be heated in oven. They are also in two sizes, one twice big (in radius) than the other. All four are put together to be heated to oven temperature. Choose the correct option from the following:

(A) Both size gulab jamuns will get heated in the same time.
(B) Smaller gulab jamuns are heated before bigger ones.
(C) Smaller pizzas are heated before bigger ones.
(D) Bigger pizzas are heated before smaller ones. **EH0119**

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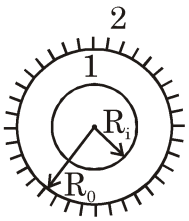
26. A calorimeter of negligible heat capacity contains 100 cc of water at 40°C. The water cools to 35°C in 300 seconds. The water is now replaced by K-oil of equal volume at 40°C. Find the time taken for the temperature to becomes 35°C under similar conditions. Specific heat capacities of water and K-oil are 4200 J/kg-K and 2100 J/kg-K respectively. Density of K-oil = 800 kg/m^3.
(A) 120 sec (B) 240 sec (C) 360 sec (D) 480 sec **EH0372**

27. An object is being heated by a heater supplying 60 W heat. Temperature of surrounding is 20°C and the temperature of object becomes constant at 50°C. Now the heater is switched off. What is the rate at which the object will lose heat when its temperature has dropped to 30°C
(A) 20 W (B) 30 W (C) 40 W (D) 60 W **EH0373**

28. Consider two long black concentric cylinders of radii R_i and R_0 as shown. They are evacuated and are radiating energy due to their own temperatures. The fraction of radiation emitted by outer cylinder that falls on itself is_____.



(A) $1-\sqrt{\frac{R_i}{R_0}}$ (B) $\sqrt{1-\frac{R_i}{R_0}}$

(C) $1-\left(\frac{R_i}{R_0}\right)^{1/3}$ (D) $1-\frac{R_i}{R_0}$

EH0374

29. A copper sphere is suspended in an evacuated chamber maintained at 300 K. The sphere is maintained at a constant temperature of 500 K by heating it electrically. A total of 210 W of electric power is needed to do it. When the surface of the copper sphere is completely blackened, 700 W is needed to maintain the same temperature of the sphere. Calculate the emissivity of copper.
(A) 0.3 (B) 0.4 (C) 0.5 (D) 0.6 **EH0375**

30. A system S receives heat continuously from an electrical heater of power 10W. The temperature of S becomes constant at 50°C when the surrounding temperature is 20°C. After the heater is switched off, S cools from 35.1°C to 34.9°C in 1 minute. The heat capacity of S is
(A) 100J/°C (B) 300J/°C (C) 750J/°C (D) 1500J/°C
EH0093

31. A heated object (at time t = 0 and temperature $T = T_0$) is taken out of the oven to cool and placed on a table near an open window. Write an expression for its temperature as function of time T, where T_S is the surrounding temperature.

(A) $T = T_s - (T_0 + T_s)e^{-kt}$ (B) $T = T_s + (T_0 + T_s)e^{-kt}$

(C) $T = T_S + (T_0 - T_S)e^{-kt}$ (D) $T = T_S - (T_0 - T_S)e^{-kt}$ **EH0376**

32. According to Wien's displacement law the wavelength of peak emission from a body is inversely proportional to the temperature of the body in kelvin. Two stars Aand B with surface temperatures 3000 kelvin and 6000 kelvin respectively, are at the same distance from the Earth. They arc of equal radius. The correct statement about the radiations emitted by the two stars is
(A) Both A and B will look equally bright in visible.
(B) A will look brighter than B in infrared.
(C) B will look brighter than A in infrared.
(D) A will emit more in the visible range than in the infrared. **EH0377**

33. A solid copper sphere (density ρ and specific heat c) of radius r at an initial temeprature 200K is suspended inside a chamber whose walls are at almost 0 K. The time required for the temeprature of the sphere to drop to 100 K is :- (Assume copper sphere as a black body.)
(A) 1.7 ρ rc (B) 2.7 ρ rc (C) 3.3 ρ rc (D) 4.2 ρ rc **EH0378**

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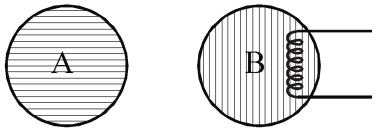
MATRIX MATCH TYPE QUESTION

Radiation

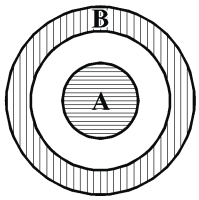
34. A & B are two black bodies of radii r_A and r_B respectively, placed in surrounding of temperature T_0. At steady state the temperature of A & B is T_A & T_B respectively.

Column I **Column II**

(A) 

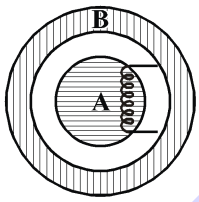
(P) $T_A = T_B$

- A & B are solid sphere
- $r_A = r_B$
- Body 'B' is being heated by a heater of constant power 'P'

(B) 

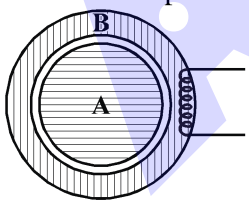
(Q) $T_A < T_B$

- B is thin spherical shell
- A is a solid sphere
- $r_A < r_B$

(C) 

(R) Heat received by A is more than heat radiated by it at steady state.

- B is thin spherical shell
- A is a solid sphere
- $r_A < r_B$
- Body A is being heated by a heater of constant power 'P'

(D) 

(S) Radiation spectrum of A & B is distinguishable

(T) Steady state can't be achieved

- B is thin spherical shell
- A is a solid sphere
- $r_A \approx r_B$
- Body B is being heated by a heater of constant power 'P'

EH0125

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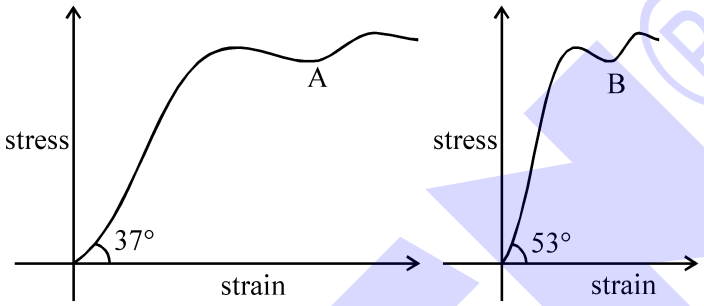
EXERCISE (S)

Elasticity

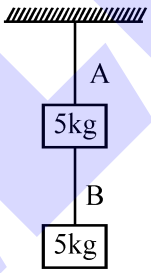
1. A steel wire of length 4.5 m and a copper wire of length 3.5 m are stretched same amount under a given load. If ratio of youngs modulli of steel to that of copper is $\frac{12}{7}$, then what is the ratio of cross sectional area of steel wire to copper wire ? **EH0001**

2. Diagram shows stress-strain graph for two material A & B. The graphs are drawn to scale.



The ratio of young modulli of material A to material B is. **EH0002**

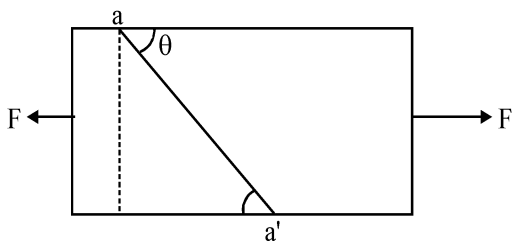
3. Two identical wires A & B of same material are loaded as shown in figure. If the elongation in wire B is 1.5 mm, what is the elongation in A. (Mass of A & B can be neglected)



EH0003

4. A wire of length L and radius r is clamped rigidly at one end. When the other end of the wire is pulled by a force f, its length increases by l. Another wire of the same material of length $2L$ and radius $2r$, is pulled by a force $2f$. Find the increase in length of this wire. **EH0004**

5. Consider a long steel bar under a tensile stress due to forces $\vec{F}$ acting at the edges along the length of the bar (Fig.). Consider a plane making an angle θ with the length. What are the tensile and shearing stresses on this plane?



(a) For what angle is the tensile stress a maximum?
(b) For what angle is the shearing stress a maximum? **EH0005**

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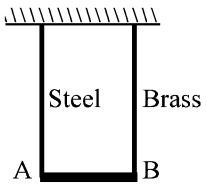
6. A light rigid bar AB is suspended horizontally from two vertical wires, one of steel and one of brass, as shown in figure. Each wire is 2.00 m long. The diameter of the steel wire is 0.60 mm and the length of the bar AB is 0.20 m. When a mass of 10 kg is suspended from the centre of AB bar remains horizontal.



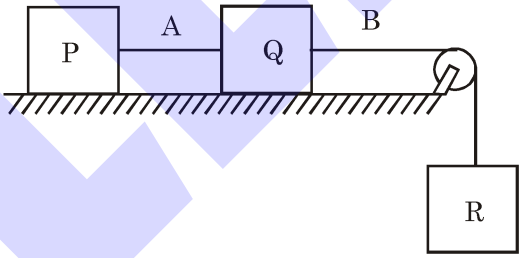
(i) What is the tension in each wire?
(ii) Calculate the extension of the steel wire and the energy stored in it.
(iii) Calculate the diameter of the brass wire.
(iv) If the brass wire were replaced by another brass wire of diameter 1 mm, where should the mass be suspended so that AB would remain horizontal? The Young modulus for steel = 2.0×10^{11} Pa, the Young modulus for brass = 1.0×10^{11} Pa. **EH0006**

7. A block of soft rubber, 5 cm cube, has one face fixed, while the opposite face is sheared through a distance 0.01 rad parallel to the fixed face by a tangential force of 40 N. How much work is done by external agent to do this. How much elastic energy is stored. **EH0379**

8. Find the elastic deformation energy of a steel rod whose one end is fixed and the other is twisted through an angle $\phi = 60°$. The length of the rod is equal to $\ell = 1.0$ m, and the radius to r = 10 mm. Shear modulus of steel is 81 GPa. **EH0380**

9. (a) A wire, 4 meters long and 0.3 mm in diameter, is stretched by a force of 800 gm wt. If the extension in length amounts to 1.5 mm, calculate the energy stored in the wire.
(b) Find the work done in joules in stretching a wire of cross section 1 sq. mm and length 2 metres through 0.1 mm if Young's modulus for the material of the wire is 2×10^{12} dynes/cm^2.
EH0381

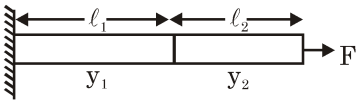
10. Each of the three blocks P, Q and R shown in figure has a mass of 3 kg. Each of the wires A and B has cross-sectional area 0.005 cm^2 and Young's modulus 2×10^{11} N/m^2. Neglect friction. Find the longitudinal strain developed in each of the wires. Take g = 10 m/s^2. **EH0382**



11. A wire of cross-sectional area 4×10^{-4} m^2, modulus of elasticity 2×10^{11} N/m^2 and length 1 m is stretched between two vertical rigid poles. A mass of 1 kg is suspended at its middle. Calculate the angle it makes with the horizontal. **EH0031**

12. Deduce an expression for the moment of the couple required to twist the lower end of a bar of circular cross section by 90°, the upper end being clamped. **EH0383**

13. Find the



(i) Net elongation of composite rod approximately. ($x \times 10^{-11}$m) then x =
(ii) Y_{eq} of the composite rod ($x \times 10^{11}$N /m^2)
(assume A = area of cross section of each rod), then x =
$\ell_1 = \ell_2 = 1$m, F = 2N/m^2 , A =1sq.m^2
$y_1 = 2 \times 10^{11}$ N/m^2 , $y_2 = 3 \times 10^{11}$ N /m^2 **EH0384**

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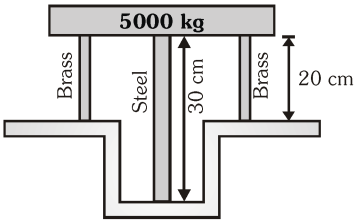
14. A steel rod of cross–sectional area 16 cm^2 and two brass rods of cross section area 10 cm^2 together support a load of 5000 kg as shown in figure. The stress in steel rod will be :
(Take Y for steel = 2.0×10^6 N/cm^2 and for brass = 1.0×10^6 N/cm^2) **EH0385**



15. Estimate the change in the density of water in ocean at a depth of 400 m below the surface. The density of water at the surface = 1030 kg/m^3 and the bulk modulus of water = 2×10^9 N/m^2.
EH0386

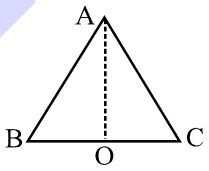
16. Find the relative change in the density of a cylindrical copper rod when it is compressed by a pressure of p = 1,000 kgf/cm^2. Poisson's ratio is $\mu = 0.34$ for copper. take $Y_{copper} = 1.18 \times 10^{11}$ N/m^2.
EH0387

17. A rubber hose 50 cm long and with an internal diameter of 1 cm is stretched until its length becomes greater by 10 cm. Find the internal diameter of the stretched hose if Poisson's ratio for rubber is 0.5.
EH0388

18. A shaft of diameter 8 cm and length 5 meters is transmitting power of 8 Kilowatts at 300 revolutions per minute. If the coefficient of rigidity of the material of the shaft be 8×10^{11} $dynes/cm^2$, find the relative shift between the ends of the shaft. **EH0389**

Thermal expansion

19. Three aluminium rods of equal length form an equilateral triangle ABC. Taking O (mid point of rod BC) as the origin. Find the increase in Y-coordinate of center of mass per unit change in temperature of the system. Assume the length of the each rod is 2m, and $\alpha_{al} = 4\sqrt{3} \times 10^{-6}$ /°C



EH0029

20. A metal rod A of 25cm lengths expands by 0.050 cm, when its temperature is raised from 0°C to 100°C. Another rod B of a different metal of length 40cm expands by 0.040 cm for the same rise in temperature. A third rod C of 50cm length is made up of pieces of rods A and B placed end to end expands by 0.03 cm on heating from 0°C to 50°C. Find the lengths of each portion of the composite rod.
EH0030

21. An isosceles triangle is formed with a rod of length l_1 and coefficient of linear expansion α_1 for the base and two thin rods each of length l_2 and coefficient of linear expansion α_2 for the two pieces, if the distance between the apex and the midpoint of the base remain unchanged as the temperatures varied show that $\frac{l_1}{l_2} = 2\sqrt{\frac{\alpha_2}{\alpha_1}}$. **EH0033**

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22. Steel wire of length 'L' at 40°C is suspended from the ceiling and then a mass 'm' is hung from its free end. The wire is cooled down from 40° to 30° C to regain its original length 'L'. The coefficient of linear thermal expansion of the steel is 10^{-5}/°C, Young's modulus of steel is 10^{11} N/m^2 and radius of the wire is 1 mm. Assume that L>> diameter of the wire. Then the value of 'm' in kg is nearly

[JEE 2011]

EH0146

23. Show that moment of inertia of a solid body of any shape changes with temperature as $I = I_0 (1 + 2\alpha\theta)$, Where I_0, is the moment of inertia at 0°C and α is the coefficient of linear expansion of the solid.

EH0390

24. A circular disc made of iron is rotated about its axis at a constant velocity ω. Calculate the percentage change in the linear speed of particle of the rim as the disc is slowly heated from 20°C to 50°C keeping the angular velocity constant. Coefficient of linear expansion of iron = 1.2×10^{-5}/°C.

EH0391

25. If two rods of length L and 2 L having coefficients of linear expansion α and 2α respectively are connected so that total length becomes 3 L, determine the average coefficient of linear expansion of the composite rod. **EH0015**

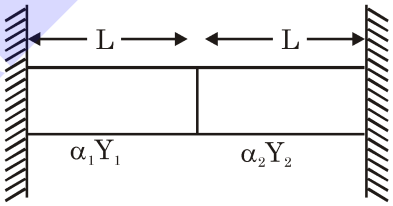
26. A clock pendulum made of invar has a period of 0.5 sec at 20°C. If the clock is used in a climate where average temperature is 30°C, approximately. How much fast or slow will the clock run in 10^6 sec. ($\alpha_{invar} = 1\times10^{-6}$/°C) **EH0016**

27. An iron bar (Young's modulus = 10^{11} N/m^2, $\alpha = 10^{-6}$ /°C) 1 m long and 10^{-3} m^2 in area is heated from 0°C to 100°C without being allowed to bend or expand. Find the compressive force developed inside the bar.

EH0017

28. A steel ball initially at a pressure of 1.0×10^5 Pa is heated from 20°C to 120°C keeping its volume constant. Find the pressure inside the ball. Coefficient of linear expansion of steel = 12×10^{-6}/°C and bulk modulus of steel = 1.6×10^{11} N/m^2. **EH0392**

29. Two metal rods are fixed end to end between two rigid supports as shown in figure. Each rod is of length ℓ and area of cross-section is A. When the system is heated up, determine the condition when the junction between rods does not shift ? (Y_1 and Y_2 are Young's modulus of materials of rods, α_1 and α_2 are coefficients of linear expansion) **EH0393**



Calorimetry

30. A steel drill making 180 rpm is used to drill a hole in a block of steel. The mass of the steel block and the drill is 180 gm. If the entire mechanical work is used up in producing heat and the rate of raise in temperature of the block and the drill is 0.5 °C/s. Find
(a) the rate of working of the drill in watts, and (b) the torque required to drive the drill.
Specific heat of steel = 0.1 and J = 4.2 J/cal. Use : $P = \tau\omega$ **EH0034**

31. An aluminium container of mass 100 gm contains 200 gm of ice at – 20°C. Heat is added to the system at the rate of 100 cal/s. Find the temperature of the system after 4 minutes (specific heat of ice = 0.5 and L = 80 cal/gm, specific heat of Al = 0.2 cal/gm/°C) **EH0008**

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32. A hot liquid contained in a container of negligible heat capacity loses temperature at rate 3 K/min, just before it begins to solidify. The temperature remains constant for 30 min. Find the ratio of specific heat capacity of liquid to specific latent heat of fusion is in K^{-1} (given that rate of losing heat is constant).

EH0009

33. Two 50 gm ice cubes are dropped into 250 gm of water into a glass. If the water was initially at a temperature of 25°C and the temperature of ice –15°C. Find the final temperature of water. (specific heat of ice = 0.5 cal/gm/°C and L = 80 cal/gm). Find final amount of water and ice.

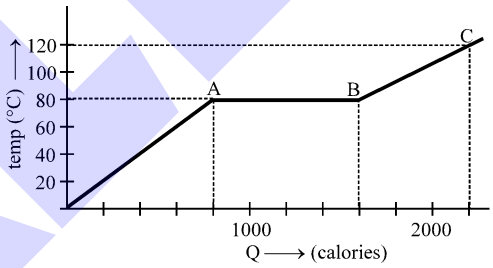
EH0010

34. A flow calorimeter is used to measure the specific heat of a liquid. Heat is added at a known rate to a stream of the liquid as it passes through the calorimeter at a known rate. Then a measurement of the resulting temperature difference between the inflow and the outflow points of the liquid stream enables us to compute the specific heat of the liquid. A liquid of density 0.2 g/cm^3 flows through a calorimeter at the rate of 10 cm^3/s. Heat is added by means of a 250-W electric heating coil, and a temperature difference of 25°C is established in steady-state conditions between the inflow and the outflow points. Find the specific heat of the liquid. **EH0011**

35. Two identical calorimeter A and B contain equal quantity of water at 20°C. A 5 gm piece of metal X of specific heat 0.2 cal g^{-1} $(C°)^{-1}$ is dropped into A and a 5 gm piece of metal Y into B. The equilibrium temperature in A is 22°C and in B 23°C. The initial temperature of both the metals is 40°C. Find the specific heat of metal Y in cal g^{-1} $(C°)^{-1}$. **EH0012**

36. The temperature of 100 gm of water is to be raised from 24° C to 90° C by adding steam to it. Calculate the mass of the steam required for this purpose. **EH0013**

37. A substance is in the solid form at 0°C. The amount of heat added to this substance and its temperature are plotted in the following graph. If the relative specific heat capacity of the solid substance is 0.5, find from the graph



(i) the mass of the substance ;

(ii) the specific latent heat of the melting process, and

(iii) the specific heat of the substance in the liquid state. **EH0014**

38. A copper calorimeter of mass 100 gm contains 200 gm of a mixture of ice and water. Steam at 100°C under normal pressure is passed into the calorimeter and the temperature of the mixture is allowed to rise to 50°C. If the mass of the calorimeter and its contents is now 330 gm, what was the ratio of ice and water in the beginning? Neglect heat losses.

Given : Specific heat capacity of copper = 0.42×10^3 J $kg^{-1}K^{-1}$,

Specific heat capacity of water = 4.2×10^3 J $kg^{-1}K^{-1}$,

Specific heat of fusion of ice = 3.36×10^5 J kg^{-1}

Latent heat of condensation of steam = 22.5×10^5 Jkg^{-1} **EH0394**

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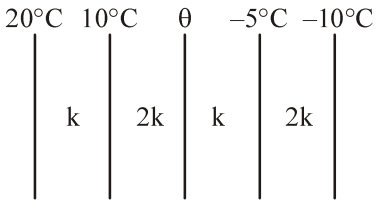
Conduction

39. The figure shows the face and interface temperature of a composite slab containing of four layers of two materials having identical thickness. Under steady state condition, find the value of temperature θ.



k = thermal conductivity

EH0019

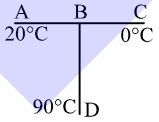
40. A thin walled metal tank of surface area $5m^2$ is filled with water and contains an immersion heater dissipating 1 kW. The tank is covered with 4 cm thick layer of insulation whose thermal conductivity is 0.2 W/m/K. The outer face of the insulation is 25°C. Find the temperature of the tank in the steady state.

EH0018

41. One end of a rod, enclosed in a thermally insulating sheath, is kept at a temperature T_1 while the other, at T_2. The rod is composed of two sections whose lengths are ℓ_1 and ℓ_2 and heat conductivity coefficients x_1 and x_2. Find the temperature of the interface. **EH0395**

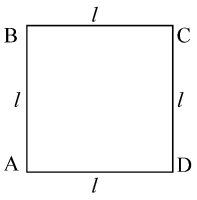
42. Two rods whose lengths are ℓ_1 and ℓ_2 and heat conductivity coefficients x_1 and x_2 are placed end to end. Find the heat conductivity coefficient of a uniform rod of length $\ell_1 + \ell_2$ whose conductivity is the same as that of the system of these two rods. The lateral surfaces of the rods are assumed to be thermally insulated. **EH0396**

43. Three conducting rods of same material and cross-section are shown in figure. Temperature of A, D and C are maintained at 20°C, 90°C and 0°C. Find the ratio of length BD and BC if there is no heat flow in AB.



EH0020

44. In the square frame of side l of metallic rods, the corners A and C are maintained at T_1 and T_2 respectively. The rate of heat flow from A to C is ω. If A and D are instead maintained T_1 & T_2 respectively. Find the total rate of heat flow.



EH0021

45. One end of copper rod of uniform cross-section and of length 1.5 meters is in contact with melting ice and the other end with boiling water. At what point along its length should a temperature of 200°C be maintained, so that in steady state, the mass of ice melting is equal to that of steam produced in the same interval of time? Assume that the whole system is insulated from the surroundings.

EH0022

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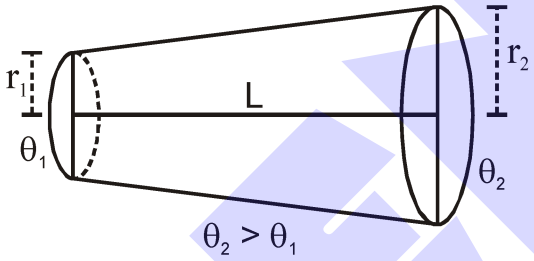
46. A cylindrical block of length 0.4 m an area of cross-section $0.04m^2$ is placed coaxially on a thin metal disc of mass 0.4 kg and of the same cross-section. The upper face of the cylinder is maintained at a constant temperature of 400K and the initial temperature of the disc is 300K. If the thermal conductivity of the material of the cylinder is 10 watt/m-K and the specific heat of the material of the disc in 600 J/kg-K, how long will it take for the temperature of the disc to increase to 350K? Assume, for purposes of calculation, the thermal conductivity of the disc to be very high and the system to be thermally insulated except for the upper face of the cylinder. **EH0038**

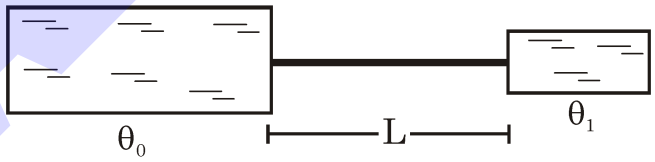
47. A rod of length L with sides fully insulated is made of a material whose thermal conductivity K varies with temperature as $K = \frac{\alpha}{T}$, where α is constant. The ends of rod are at temperature T_1 and T_2 $(T_2 > T_1)$. Find the rate of heat flow per unit area of rod. **EH0397**

48. Find the rate of heat flow through a cross-section of the rod shown in figure $(\theta_2 > \theta_1)$. Thermal conductivity of the material of the rod is K. **EH0398**



49. A hollow tube has a length l, inner radius R_1 and outer radius R_2. The material has a thermal conductivity K. Find the heat flowing through the walls of the tube if
[a] the flat ends are maintained at temperatures T_1 and $T_2(T_2 > T_1)$
[b] the inside of the tube is maintained at temperature T_1 and the outside is maintained at temperature T_1 and the outside is maintained at T_2. **EH0399**

50. Figure shows a large tank of water at a constant temperature è0 and a small vessel containing a mass m of water at an initial temperature $\theta_1 (< \theta_0)$. A metal rod of length L, area of cross section A and thermal conductivity K connects the two vessels. Find the time taken for the temperature of the water in the smaller vessel to become $\theta_2 (\theta_1 < \theta_2 < \theta_0)$. Specific heat capacity of water is s and all other heat capacities are negligible. **EH0400**



Radiation

51. A solid aluminium sphere and a solid copper sphere of twice the radius are heated to the same temperature and are allowed to cool under identical surrounding temperatures. Assume that the emissivity of both the spheres is the same. Find the ratio of
[a] the rate of heat loss from the aluminium sphere to the rate of heat loss from the copper sphere and
[b] the rate of fall of temperature of the aluminium sphere to the rate of fall of temperature of the copper sphere. The specific heat capacity of aluminium = 900 J/kg–°C and that of copper = 390 J/kg-°C. The density of copper = 3.4 times the density of aluminium. **EH0401**

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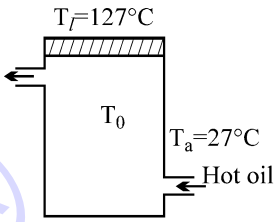
52. A spherical ball A of surface area 20 cm^2 is kept at the centre of a hollow spherical shell B of area 80 cm^2. The surface of A and the inner surface of B emit as blackbodies. Assume that the thermal conductivity of the material of B is very poor and that of A is very high and that the air between A and B has been pumped out. The heat capacities of A and B are 42 J/°C and 82 J/°C respectively. Initially, the temperature of A is 100°C and that of B is 20°C. Find the rate of change of temperature of A and that of B at this instant. Explain the effects of the asumptions listed in the problem. **EH0402**

53. Hot oil is circulated through an insulated container with a wooden lid at the top whose conductivity K = 0.149 J/(m-°C-sec), thickness t = 5 mm, emissivity = 0.6. Temperature of the top of the lid in steady state is at T_l = 127°. If the ambient temperature T_a = 27°C. Calculate **[JEE' 2003]**



(a) rate of heat loss per unit area due to radiation from the lid.

(b) temperature of the oil. (Given $\sigma = \frac{17}{3}\times10^{-8}$)

EH0040

54. Two spheres of same radius R have their densities in the ratio 8 : 1 and the ratio of their specific heats are 1 : 4. If by radiation their rates of fall of temperature are same, then find the ratio of their rates of losing heat. **EH0023**

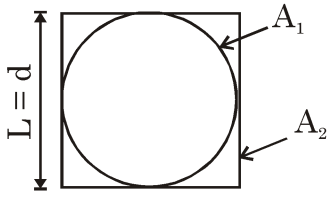
55. A solid receives heat by radiation over its surface at the rate of 4 kW. The heat convection rate from the surface of solid to the surrounding is 5.2 kW, and heat is generated at a rate of 1.7 kW over the volume of the solid. The rate of change of the average temperature of the solid is 0.5°Cs^{-1}. Find the heat capacity of the solid. **EH0024**

56. A solid copper cube and sphere, both of same mass & emissivity are heated to same initial temperature and kept under identical conditions. What is the ratio of their initial rate of fall of temperature? **EH0025**

57. The earth receives solar radiation at a rate of 8.2 J cm^{-2} min^{-1}. Assuming that the sun radiates like a blackbody, calculate the surface temperature of the sun. The angle subtended by the sun on the earth is 0.53 and the Stefan constant $\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4}. **EH0403**

58. Derive an expression for the blackbody temperature of a planet with radius r and albedo α, orbiting its sun at a certain distance D. The star has a temperature T. and radius R. Albedo is fraction of light that surface reflects. Although the body is blackbody albedo is due to atmosphere, ocean etc. **EH0404**

59. A sphere of diameter L is enclosed inside a cube of side L as shown. Assume that the side of the cube is slightly greater than diameter so that they do not touch each other. The region between them is evacuated. The bodies are painted black. If they are at a temperature T_1 and T_2 respectively, what is the rate of heat exchange between them? **EH0405**



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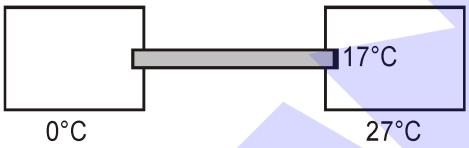
60. One end of a rod of length 20 cm is inserted in a furnace at 800 K. The sides of the rod are covered with an insulating material and the other end emits radiation like a blackbody. The temperature of this end is 750 k in the steady state. The temperature of the surrounding air is 300 K. Assuming radiation to be the only important mode of energy transfer between the surrounding and the open end of the rod, find the thermal conductivity of the rod. Stefan constant $\sigma = 6.0 \times 10^{-8}$ W/m^2–K^4

EH0406

61. A cylindrical rod of length 50 cm and cross-sectional area 1 cm^2. is fitted between a large ice chamber at 0°C and an evacuated chamber maintained at 27°C as shown in figure. Only small portions of the rod are inside the chambers and the rest is thermally insulated from the surrounding. The cross-section going into the evacuated chamber is blackened so that it completely absorbs any radiation falling on it. The temperature of the blackened end is 17°C when steady state is reached. Stefan constant $\sigma = 6 \times 10^{-8}$ W/m^2-K^4. Find the thermal conductivity of the material of the rod.

EH0407



62. The maximum in the energy distribution spectrum of the sun is at 4753 Å and its temperature is 6050K. What will be the temperature of the star whose energy distribution shows a maximum at 9506Å.

EH0027

63. A liquid takes 5 minutes to cool from 80°C to 50°C. How much time will it take to cool from 60°C to 30°C ? The temperature of surrounding is 20°C. Use exact method.

EH0039

64. A vessel containing 100 gm water at 0°C is suspended in the middle of a room. In 15 minutes the temperature of the water rises by 2°C. When an equal amount of ice is placed in the vessel, it melts in 10 hours. Calculate the specific heat of fusion of ice.

EH0026

65. One day in the morning, Ramesh filled up 1/3 bucket of hot water from geyser, to take bath. Remaining 2/3 was to be filled by cold water (at room temperature) to bring mixture to a comfortable temperature. Suddenly Ramesh had to attend to something which would take some times, say 5-10 minutes before he could take bath. Now he had two options: (i) fill the remaining bucket completely by cold water and then attend to the work, (ii) first attend to the work and fill the remaining bucket just before taking bath. Which option do you think would have kept water warmer ? Explain.

EH0007

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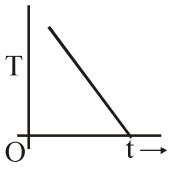
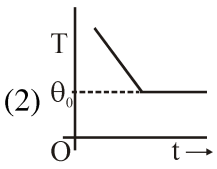
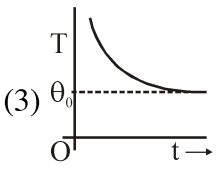
EXERCISE (JM)

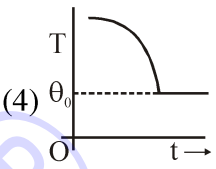
1. If a piece of metal is heated to temperature θ and then allowed to cool in a room which is at temperature θ_0 the graph between the temperature T of the metal and time t will be closed to:

[JEE-Main- 2013]

(1) 

(2) 

(3) 

(4) 

EH0132

2. The pressure that has to be applied to the ends of a steel wire of length 10 cm to keep its length constant when its temperature is raised by 100°C is :
(For steel Young's modulus is 2×10^{11} N m^{-2} and coefficient of thermal expansion is 1.1×10^{-5} K^{-1})

[JEE-Main-2014]

(1) 2.2×10^7 Pa (2) 2.2×10^6 Pa (3) 2.2×10^8 Pa (4) 2.2×10^9 Pa

EH0133

3. Three rods of Copper, Brass and Steel are welded together to form a Y-shaped structure. Area of cross-section of each rod = 4 cm^2. End of copper rod is maintained at 100°C where as ends of brass and steel are kept at 0°C. Lengths of the copper, brass and steel rods are 46, 13 and 12 cms respectively. The rods are thermally insulated from surroundings except at ends. Thermal conductivities of copper, brass and steel are 0.92, 0.26 and 0.12 CGS units respectively. Rate of heat flow through copper rod is :

[JEE-Main-2014]

(1) 4.8 cal/s (2) 6.0 cal/s (3) 1.2 cal/s (4) 2.4 cal/s

EH0134

4. A pendulume clock loses 12s a day if the temperature is 40°C and gains 4s a day if the temperature is 20°C. The temperature at which the clock will show correct time, and the coeffecient of linear expansion (α) of the metal of the pendulum shaft are respectively :- **[JEE-Main-2016]**

(1) 55°C ; $\alpha = 1.85 \times 10^{-2}$ / °C (2) 25°C ; $\alpha = 1.85 \times 10^{-5}$ / °C
(3) 60°C ; $\alpha = 1.85 \times 10^{-4}$ / °C (4) 30°C ; $\alpha = 1.85 \times 10^{-3}$ / °C

EH0136

5. A copper ball of mass 100 gm is at a temperature T. It is dropped in a copper calorimeter of mass 100gm, filled with 170 gm of water at room temperature. Subsequently, the temperature of the system is found to be 75°C. T is given by : (Given : room temperature = 30° C, specific heat of copper = 0.1 cal/gm°C) **[JEE-Main-2017]**

(1) 1250°C (2) 825°C (3) 800°C (4) 885° C

EH0137

6. A man grows into a giant such that his linear dimensions increase by a factor of 9. Assuming that his density remains same, the stress in the leg will change by a factor of : **[JEE-Main-2017]**

(1) 81 (2) $\frac{1}{81}$ (3) 9 (4) $\frac{1}{9}$

EH0138

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7. An external pressure P is applied on a cube at 0°C so that it is equally compressed from all sides. K is the bulk modulus of the material of the cube and α is its coefficient of linear expansion. Suppose we want to bring the cube to its original size by heating. The temperature should be raised by : **[JEE-Main-2017]**

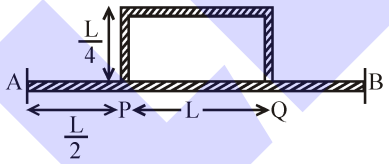
(1) $\frac{3\alpha}{PK}$ (2) $3PK\alpha$ (3) $\frac{P}{3\alpha K}$ (4) $\frac{P}{\alpha K}$ **EH0139**

8. A solid sphere of radius r made of a soft material of bulk modulus K is surrounded by a liquid in a cylindrical container. A massless piston of area a floats on the surface of the liquid, covering entire cross section of cylindrical container. When a mass m is placed on the surface of the piston to compress the liquid, the fractional decrement in the radius of the sphere, $\left(\frac{dr}{r}\right)$, is : **[JEE-Main-2018]**

(1) $\frac{Ka}{3mg}$ (2) $\frac{mg}{3Ka}$ (3) $\frac{mg}{Ka}$ (4) $\frac{Ka}{mg}$ **EH0140**

SELECTED PROBLEMS FROM JEE-MAINS ONLINE PAPERS

9. Temperature difference of 120°C is maintained between two ends of a uniform rod AB of length 2L. Another bent rod PQ, of same cross-section as AB and length $\frac{3L}{2}$, is connected across AB (See figure). In steady state, temperature difference between P and Q will be close to : **[JEE-Main-2019_Jan]**



(1) 60ºC (2) 75ºC (3) 35ºC (4) 45ºC **EH0408**

10. An unknown metal of mass 192 g heated to a temperature of 100ºC was immersed into a brass calorimeter of mass 128 g containing 240 g of water a temperature of 8.4ºC Calculate the specific heat of the unknown metal if water temperature stabilizes at 21.5ºC (Specific heat of brass is 394 J kg^{-1} K^{-1}) **[JEE-Main-2019_Jan]**

(1) 1232 J kg^{-1} K^{-1} (2) 458 J kg^{-1} K^{-1} (3) 654 J kg^{-1} K^{-1} (4) 916 J kg^{-1} K^{-1}

EH0409

11. A thermometer graduated according to a linear scale reads a value x_0 when in contact with boiling water, and $x_0/3$ when in contact with ice. What is the temperature of an object in 0 ºC, if this thermometer in the contact with the object reads $x_0/2$? **[JEE-Main-2019_Jan]**

(1) 35 (2) 25 (3) 60 (4) 40 **EH0410**

12. A cylinder of radius R is surrounded by a cylindrical shell of inner radius R and outer radius 2R. The thermal conductivity of the material of the inner cylinder is K_1 and that of the outer cylinder is K_2. Assuming no loss of heat, the effective thermal conductivity of the system for heat flowing along the length of the cylinder is: **[JEE-Main-2019_Jan]**

(1) $K_1 + K_2$ (2) $\frac{K_1 + K_2}{2}$ (3) $\frac{2K_1 + 3K_2}{5}$ (4) $\frac{K_1 + 3K_2}{4}$

EH0411

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13. A boy's catapult is made of rubber cord which is 42 cm long, with 6 mm diameter of cross-section and of negligible mass. The boy keeps a stone weighing 0.02kg on it and stretches the cord by 20 cm by applying a constant force. When released, the stone flies off with a velocity of 20 ms^{-1}. Neglect the change in the area of cross-section of the cord while stretched. The Young's modulus of rubber is closest to: **[JEE-Main-2019_April]**

(1) 10^4 Nm^{-2} (2) 10^8 Nm^{-2} (3) 10^6 Nm^{-2} (4) 10^3 Nm^{-2} **EH0412**

14. A thermally insulated vessel contains 150g of water at 0°C. Then the air from the vessel is pumped out adiabatically. A fraction of water turns into ice and the rest evaporates at 0°C itself. The mass of evaporated water will be closest to :

(Latent heat of vaporization of water = 2.10×10^6 J kg^{-1} and Latent heat of Fusion of water = 3.36×10^5 J kg^{-1}) **[JEE-Main-2019_April]**

(1) 130 g (2) 35 g (3) 20 g (4) 150 g **EH0413**

15. The elastic limit of brass is 379 MPa. What should be the minimum diameter of a brass rod if it is to support a 400 N load without exceeding its elastic limit ? **[JEE-Main-2019_April]**

(1) 1.16 mm (2) 0.90 mm (3) 1.36 mm (4) 1.00 mm **EH0414**

16. A non-isotropic solid metal cube has coefficients of linear expansion as :

5×10^{-5}/°C along the x-axis and 5×10^{-6}/°C along the y and the z-axis. If the coefficient of volume expansion of the solid is $C \times 10^{-6}$/°C then the value of C is _____. **[JEE-Main-2020_Jan]**

EH0415

17. M grams of steam at 100°C is mixed with 200 g of ice at its melting point in a thermally insulated container. If it produces liquid water at 40°C [heat of vaporization of water is 540 cal/ g and heat of fusion of ice is 80 cal/g], the value of M is_____. **[JEE-Main-2020_Jan]**

EH0416

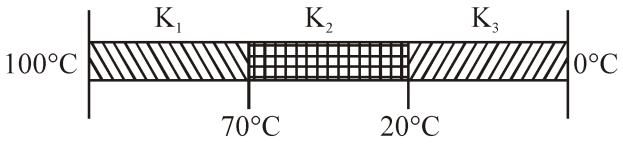
18. A bakelite beaker has volume capacity of 500 cc at 30°C. When it is partially filled with V_m volume (at 30°) of mercury, it is found that the unfilled volume of the beaker remains constant as temperature is varied. If $\gamma_{(beaker)} = 6 \times 10^{-6}$ $°C^{-1}$ and $\gamma_{(mercury)} = 1.5 \times 10^{-4}$ $°C^{-1}$, where γ is the coefficient of volume expansion, then V_m (in cc) is close to_____. **[JEE-Main-2020_Sep]**

EH0417

19. A bullet of mass 5g, travelling with a speed of 210 m/s, strikes a fixed wooden target. One half of its kinetic energy is converted into heat in the bullet while the other half is converted into heat in the wood. The rise of temperature of the bullet if the specific heat of its material is 0.030 cal/(g–°C) (1 cal = 4.2×10^7 ergs) close to : **[JEE-Main-2020_Sep]**

(1) 83.3°C (2) 87.5°C (3) 119.2°C (4) 38.4°C **EH0418**

20. Three rods of identical cross-section and lengths are made of three different materials of thermal conductivity K_1, K_2, and K_3, respectively. They are joined together at their ends to make a long rod (see figure). One end of the long rod is maintained at 100°C and the other at 0°C (see figure). If the joints of the rod are at 70°C and 20°C in steady state and there is no loss of energy from the surface of the rod, the correct relationship between K_1, K_2 and K_3 is : **[JEE-Main-2020_Sep]**



(1) $K_1 : K_3 = 2 : 3$; $K_2 : K_3 = 2 : 5$ (2) $K_1 < K_2 < K_3$

(3) $K_1 : K_2 = 5 : 2$; $K_1 : K_3 = 3 : 5$ (4) $K_1 > K_2 > K_3$ **EH0419**

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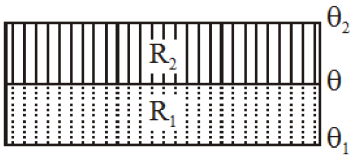
21. The temperature θ at the junction of two insulating sheets, having thermal resistances R_1 and R_2 as well as top and bottom temperatures θ_1 and θ_2 (as shown in figure) is given by :

[JEE-Main-2021_Feb]



(1) $\frac{\theta_2 R_2 - \theta_1 R_1}{R_2 - R_1}$ (2) $\frac{\theta_1 R_2 - \theta_2 R_1}{R_2 - R_1}$ (3) $\frac{\theta_1 R_2 + \theta_2 R_1}{R_1 + R_2}$ (4) $\frac{\theta_1 R_1 + \theta_2 R_2}{R_1 + R_2}$

EH0420

22. In 5 minutes, a body cools from 75°C to 65°C at room temperature of 25°C. The temperature of body at the end of next 5 minutes is _____°C. **[JEE-Main-2021_July]**

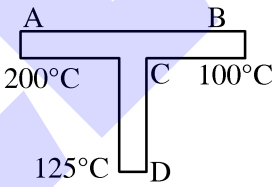
EH0421

23. The temperature of equal masses of three different liquids x,y and z are 10ºC, 20ºC and 30ºC respectively. The temperature of mixture when x is mixed with y is 16ºC and that when y is mixed with z is 26ºC. The temperature of mixture when x and z are mixed will be :

[JEE-Main-2021_Aug]

(1) 28.32º C (2) 25.62º C (3) 23.84ºC (4) 20.28ºC **EH0422**

24. A rod CD of thermal resistance 10.0 KW^{-1} is joined at the middle of an identical rod AB as shown in figure, The end A, B and D are maintained at 200°C, 100°C and 125°C respectively. The heat current in CD is P watt. The value of P is ..... . **[JEE-Main-2021_Aug]**



EH0423

25. Two thin metallic spherical shells of radii r_1 and r_2 ($r_1 < r_2$) are placed with their centres coinciding. A material of thermal conductivity K is filled in the space between the shells. The inner shell is maintained at temperature θ_1 and the outer shell at temperature θ_2($\theta_1 < \theta_2$). The rate at which heat flows radially through the material is :- **[JEE-Main-2021_Aug]**

(1) $\frac{4\pi K r_1 r_2 (\theta_2 - \theta_1)}{r_2 - r_1}$ (2) $\frac{\pi r_1 r_2 (\theta_2 - \theta_1)}{r_2 - r_1}$ (3) $\frac{K(\theta_2 - \theta_1)}{r_2 - r_1}$ (4) $\frac{K(\theta_2 - \theta_1)(r_2 - r_1)}{4\pi r_1 r_2}$

EH0424

26. A steel rod with $y = 2.0 \times 10^{11}$ Nm^{-2} and $\alpha = 10^{-5}$ $°C^{-1}$ of length 4 m and area of cross-section 10 cm^2 is heated from 0° C to 400°C without being allowed to extend. The tension produced in the rod is $x \times 10^5$ N where the value of x is ..... **[JEE-Main-2021_Aug]**

EH0425

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\3-Calorimetry, Elasticity, T.E., H.T. KTG\Eng\01_Elasticity,Heat\01_Eng.p65

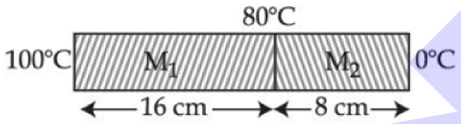
27. If the length of a wire is made double and radius is halved of its respective values. Then, the Young's modules of the material of the wire will : **[JEE-Main-2022_July]**

(1) Remains same (2) Become 8 times its initial value

(3) Become $\frac{1^{th}}{4}$ of its initial value (4) Become 4 times its initial value **EH0426**

28. A string of area of cross-section 4 mm^2 and length 0.5 is connected with a rigid body of mass 2 kg. The body is rotated in a vertical circular path of radius 0.5 m. The body acquires a speed of 5 m/s at the bottom of the circular path. Strain produced in the string when the body is at the bottom of the circle is …… $\times 10^{-5}$. (Use Young's modulus 10^{11} N/m^2 and g = 10 m/s^2) **[JEE-Main-2022_July]**
EH0427

29. Two metallic blocks M_1 and M_2 of same area of cross-section are connected to each other (as shown in figure). If the thermal conductivity of M_2 is K then the thermal conductivity of M_1 will be : [Assume steady state heat conduction] **[JEE-Main-2022_June]**

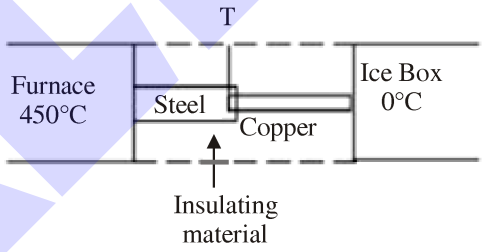


(1) 10 K (2) 8 K (3) 12.5 K (4) 2 K **EH0428**

30. A solid metallic cube having total surface area 24 m^2 is uniformly heated. If its temperature is increased by 10°C, calculate the increase in volume of the cube. (Given $\alpha = 5.0 \times 10^{-4}$ $°C^{-1}$)

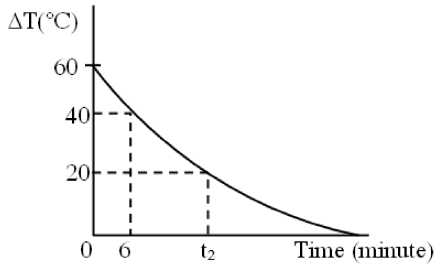
(1) 2.4×10^6 cm^3 (2) 1.2×10^5 cm^3 (3) 6.0×10^4 cm^3 (4) 4.8×10^5 m^3
[JEE-Main-2022_June]
EH0429

31. If K_1 and K_2 are the thermal conductivities L_1 and L_2 are the lengths and A_1 and A_2 are the cross sectional areas of steel and copper rods respectively such that $\frac{K_2}{K_1} = 9$, $\frac{A_1}{A_2} = 2$, $\frac{L_1}{L_2} = 2$. Then, for the arrangement as shown in the figure. The value of temperature T of the steel- copper junction in the steady state will be : **[JEE-Main-2022_July]**
EH0430



(1) 18ºC (2) 14 ºC (3) 45 ºC (4) 150 ºC

32. In an experment to verify Newton's law of cooling, a graph is plotted between, the temperature difference (ΔT) of the water and surroundings and time as shown in figure. The initial temperature of water is taken as 80°C. The value of t_2 as mentioned in the graph will be _____. **[JEE-Main-2022_June]**



EH0431

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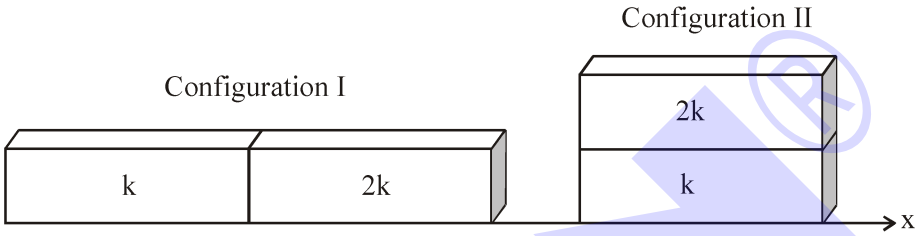
EXERCISE (JA)

1. Two rectangular blocks, having identical dimensions, can be arranged either in configuration I or in configuration II as shown in the figure. One of the blocks has thermal conductivity k and the other 2k. The temperature difference between the ends along the x-axis is the same in both the configurations. It takes 9s to transport a certain amount of heat from the hot end to the cold end in the configuration I. The time to transport the same amount of heat in the configuration II is :- **[JEE-Advance-2013]**



(A) 2.0 s (B) 3.0 s (C) 4.5 s (D) 6.0 s

EH0148

2. One end of a horizontal thick copper wire of length 2L and radius 2R is welded to an end of another horizontal thin copper wire of length L and radius R. When the arrangement is stretched by applying forces at two ends, the ratio of the elongation in the thin wire to that in the thick wire is :-

[JEE-Advance-2013]

(A) 0.25 (B) 0.50 (C) 2.00 (D) 4.00

EH0149

3. Parallel rays of light of intensity $I = 912\ Wm^{-2}$ are incident on a spherical black body kept in surroundings of temperature 300 K. Take Stefan-Boltzmann constant $\sigma = 5.7 \times 10^{-8}\ Wm^{-2}\ K^{-4}$ and assume that the energy exchange with the surroundings is only through radiation. The final steady state temperature of the black body is close to :- **[JEE-Advance-2014]**

(A) 330 K (B) 660 K (C) 990 K (D) 1550 K

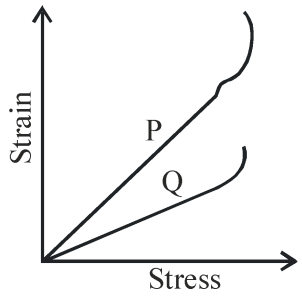
EH0150

4. Two spherical stars A and B emit blackbody radiation. The radius of A is 400 times that of B and A emits 10^4 times the power emitted from B. The ratio $\left(\frac{\lambda_A}{\lambda_B}\right)$ of their wavelengths λ_A and λ_B at which the peaks occur in their respective radiation curves is. **[JEE-Advance-2015]**

EH0151

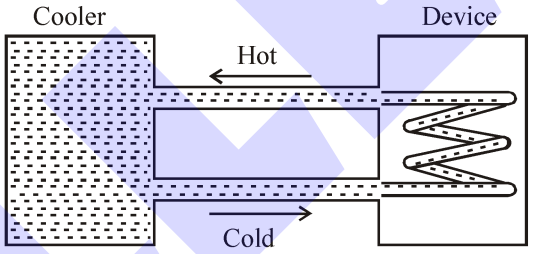
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\3-Calorimetry, Elasticity, T.E., H.T., KTG\Eng\01_Elasticity,Heat\01_Eng.p65

5. In plotting stress versus strain curves for two materials P and Q, a student by mistake puts strain on the y-axis and stress on the x-axis as shown in the figure. Then the correct statement(s) is (are):- **[JEE-Advance-2015]**



(A) P has more tensile strength than Q.
(B) P is more ductile than Q.
(C) P is more brittle than Q.
(D) The Young's modulus of P is more than that of Q. **EH0152**

6. A water cooler of storage capacity 120 litres can cool water at constant rate of P watts. In a closed circulation system (as shown schematically in the figure), the water from the cooler is used to cool an external device that generates constantly 3 kW of heat (thermal load). The temperature of water fed into the device cannot exceed 30°C and the entire stored 120 litres of water is initially cooled to 10°C. The entire system is thermally insulated. The minimum value of P (in watts) for which the device can be operated for 3 hours is : **[JEE-Advance-2016]**



(Specific heat of water is 4.2 kJ kg^{-1} K^{-1} and the density of water is 1000 kg m^{-3})
(A) 1600 (B) 2067 (C) 2533 (D) 3933 **EH0153**

7. A metal is heated in a furnace where a sensor is kept above the metal surface to read the power radiated (P) by the metal. The sensor has a scale that displays $\log_2(P/P_0)$, where P_0 is a constant. When the metal surface is at a temperature of 487 °C, the sensor shows a value 1. Assume that the emissivity of the metallic surface remains constant. What is the value displayed by the sensor when the temperature of the metal surface is raised to 2767 °C ? **[JEE-Advance-2016]**
EH0154

8. The ends Q and R of two thin wires, PQ and RS, are soldered (joined) together. Initially each of the wires has a length of 1m at 10°C. Now the end P is maintained at 10°C, while the end S is heated and maintained at 400°C. The system is thermally insulated from its surroundings. If the thermal conductivity of wire PQ is twice that of the wire RS and the coefficient of linear thermal expansion of PQ is 1.2×10^{-5} K^{-1}, the change in length of the wire PQ is **[JEE-Advance-2016]**
(A) 0.78 mm (B) 0.90 mm (C) 1.56 mm (D) 2.34 mm
EH0155

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9. A human body has a surface area of approximately 1 m^2. The normal body temperature is 10 K above the surrounding room temperature T_0. Take the room temperature to be $T_0 = 300$ K. For $T_0 = 300$ K, the value of $\sigma T_0^4 = 460\ Wm^{-2}$ (where σ is the Stefan-Boltzmann constant). Which of the following options is/are **correct** ? **[JEE-Advance-2017]**

(A) The amount of energy radiated by the body in 1 second is close to 60 Joules

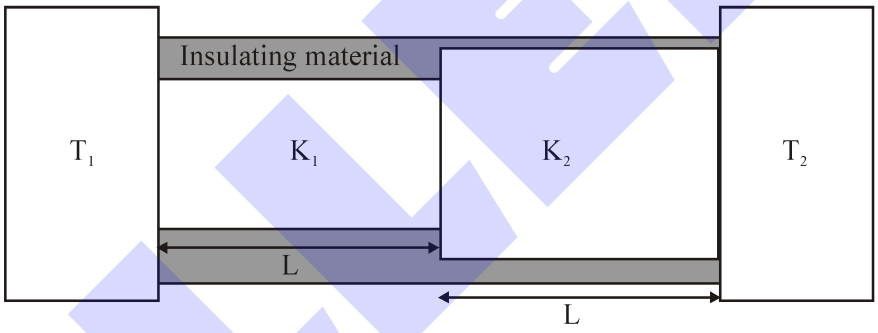
(B) If the surrounding temperature reduces by a small amount $\Delta T_0 << T_0$, then to maintain the same body temperature the same (living) human being needs to radiate $\Delta W = 4\sigma T_0^3 \Delta T_0$ more energy per unit time

(C) Reducing the exposed surface area of the body (e.g. by curling up) allows humans to maintain the same body temperature while reducing the energy lost by radiation

(D) If the body temperature rises significantly then the peak in the spectrum of electromagnetic radiation emitted by the body would shift to longer wavelengths

EH0156

10. Two conducting cylinders of equal length but different radii are connected in series between two heat baths kept at temperatures $T_1 = 300$ K and $T_2 = 100$ K, as shown in the figure. The radius of the bigger cylinder is twice that of the smaller one and the thermal conductivities of the materials of the smaller and the larger cylinders are K_1 and K_2 respectively. If the temperature at the junction of the two cylinders in the steady state is 200 K, then $K_1/K_2 =$ _____. **[JEE-Advance-2018]**

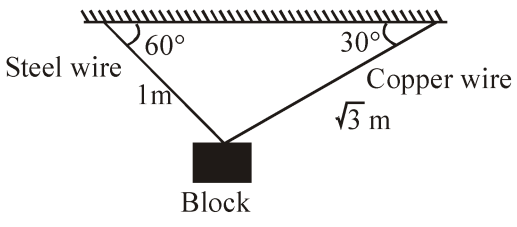


EH0157

11. A block of weight 100 N is suspended by copper and steel wires of same cross sectional area 0.5 cm^2 and, length $\sqrt{3}$ m and 1 m, respectively. Their other ends are fixed on a ceiling as shown in figure. The angles subtended by copper and steel wires with ceiling are 30° and 60°, respectively. If elongation in copper wire is $(\Delta\ell_C)$ and elongation in steel wire is $(\Delta\ell_S)$, then the ratio $\frac{\Delta\ell_C}{\Delta\ell_S}$ is _____.

[Young's modulus for copper and steel are 1×10^{11} N/m^2 and 2×10^{11} N/m^2 respectively]

[JEE-Advance-2019]



EH0158

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\3-Calorimetry, Elasticity, T.E., H.T. KTG\Eng\01_Elasticity,Heat\01_Eng.p65

12. A liquid at 30°C is poured very slowly into a Calorimeter that is at temperature of 110°C. The boiling temperature of the liquid is 80°C. It is found that the first 5 gm of the liquid completely evaporates. After pouring another 80 gm of the liquid the equilibrium temperature is found to be 50°C. The ratio of the Latent heat of the liquid to its specific heat will be _____ °C.
[Neglect the heat exchange with surrounding] **[JEE-Advance-2019]**

EH0159

13. The filament of a light bulb has surface area 64 mm^2. The filament can be considered as a black body at temperature 2500 K emitting radiation like a point source when viewed from far. At night the light bulb is observed from a distance of 100 m. Assume the pupil of the eyes of the observer to be circular with radius 3 mm. Then **[JEE-Advance-2020]**
(Take Stefan-Boltzmann constant = 5.67×10^{-8} Wm^{-2} K^{-4}, Wien's displacement constant = 2.90×10^{-3} m-K, Planck's constant = 6.63×10^{-34} Js, speed of light in vacuum= 3.00×10^8 ms^{-1})-
(A) power radiated by the filament is in the range 642 W to 645 W
(B) radiated power entering into one eye of the observer is in the range 3.15×10^{-8} W to 3.25×10^{-8} W
(C) the wavelength corresponding to the maximum intensity of light is 1160 nm
(D) taking the average wavelength of emitted radiation to be 1740 nm, the total number of photons entering per second into one eye of the observer is in the range 2.75×10^{11} to 2.85×10^{11}

EH0432

14. A container with 1 kg of water in it is kept in sunlight, which causes the water to get warmer than the surroundings. The average energy per unit time per unit area received due to the sunlight is $700 Wm^{-2}$ and it is absorbed by the water over an effective area of 0.05 m^2. Assuming that the heat loss from the water to the surroundings is governed by Newton's law of cooling, the difference (in °C) in the temperature of water and the surroundings after a long time will be _____.
(Ignore effect of the container, and take constant for Newton's law of cooling = 0.001 s^{-1}, Heat capacity of water = 4200 J kg^{-1} K^{-1}) **[JEE-Advance-2020]**

EH0433

15. A small object is placed at the center of a large evacuated hollow spherical container. Assume that the container is maintained at 0 K. At time t = 0, the temperature of the object is 200 K. The temperature of the object becomes 100 K at $t = t_1$ and 50 K at $t = t_2$. Assume the object and the container to be ideal black bodies. The heat capacity of the object does not depend on temperature. The ratio (t_2/t_1) is____.
[JEE-Advance-2021]

EH0434

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\3-Calorimetry, Elasticity, T.E., H.T, KTG\Eng\01_Elasticity-Heat\01_Eng.p65

ANSWER KEY

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B	B	A	A	C	B	A	A	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	B	C	A	B	D	B	A	D	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	D	D	B	A	A	C	A	A	A
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	C	A	A	D	D	D	D	A	B	B
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	B	A	D	B	C	A	A	A	A	C
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	D	A	C	B	B	D	C	B	C	A
Que.	61	62	63	64	65	66	67	68	69	70
Ans.	C	A	A	A	A	B	A	C	C	B
Que.	71	72	73	74	75	76	77	78	79	80
Ans.	D	B	D	B	C	D	A	A	A	B
Que.	81	82	83	84	85	86	87	88	89	90
Ans.	D	C	A	B	B	C	A	D	B	C
Que.	91	92	93	94	95	96	97	98	99	100
Ans.	A	A	D	B	B	C	A	B	A	A
Que.	101	102	103	104	105	106	107	108	109	110
Ans.	D	A	D	C	A	C	A	D	C	D
Que.	111	112	113	114	115	116	117	118	119	120
Ans.	B	B	D	A	A	A	A	B	B	A
Que.	121	122	123	124	125	126	127	128	129	130
Ans.	D	A	B	B	D	A	C	D	B	A
Que.	131	132	133	134	135	136	137	138	139	140
Ans.	A	B	A	A	C	A	B	A	A	B
Que.	141	142	143	144	145	146	147	148	149	150
Ans.	A	B	C	C	D	B	A	B	B	C
Que.	151	152	153	154	155	156	157	158	159	160
Ans.	C	D	D	B	B	D	D	B	C	B
Que.	161	162	163	164	165	166	167	168	169	170
Ans.	C	A	C	B	D	B	B	A,D	A,C,D	C,D
Que.	171	172	173	174	175	176	177	178	179	180
Ans.	B,C	A,C,D	B,C	A,B	B,D	A,B	A,D	B,C	C,D	A,C
Que.	181	182	183	184	185	186	187	188	189	190
Ans.	A,B	A,C	A,B	A,C	C,D	A	B	B	A,B	A,B,C,D
Que.	191									
Ans.	A-Q; B-P,R; C-Q,S; D-R									

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EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	C	B	A	A	A	A	D	B,C	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B,D	C	C	A	B	D	D	D	B	B,C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	B	A,C,D or A,B,C,D	B	B	A	A	D	A	D
Que.	31	32	33	34						
Ans.	C	C	A	A-Q,S; B-P; C-S; D-P						

EXERCISE # S

1. 0.75 **2.** $\frac{9}{16}$ **3.** 3mm **4.** ℓ' **5.** (a) $\theta = \frac{\pi}{2}$ (b) $\theta = \pi/4$.

6. (i) 50 N, (ii) 0.045 J, 1.8×10^{-3} m (iii) 8.4×10^{-4} m, (iv) x = 0.12 m **7.** 0.02J, 0.01 J

8. 697.6 J **9. (a)** 5.88×10^{4} ergs; **(b)** 5×10^{-4} joules.

10. Strain in wire A = 10^{-4} and strain in wire B = 2×10^{-4} **11.** 1/200 rad

12. $n\pi^2R^4/4L$ **13. (i)** 1.67, **(ii)** 2.4 **14.** (161.2 Kg/cm^2) **15.** 2 kg/m^3

16. 0.027% **17.** 9 mm **18.** 3.96 mrad **19.** 4×10^{-6} m/°C **20.** 10cm , 40cm

22. 3 **24.** 3.6×10^{-2} **25.** $5\alpha/3$ **26.** 5 sec slow **27.** 10, 000 N **28.** 5.8×10^{8} Pa

29. $\alpha_1 Y_1 = \alpha_2 Y_2$ **30. (a)** 37.8 J/s (Watts); **(b)** 2.005 N-m **31.** 25.5°C **32.** 1/90

33. 0 °C, 125/4 g ice, 1275/4 g water **34.** 5000 J/°C kg **35.** 27/85 **36.** 12 gm

37. (i) 0.02kg, **(ii)** 40,000calkg^{-1}, **(iii)** 750calkg^{-1}K^{-1} **38.** 1 : 1.26 **39.** 5°C **40.** 65°C

41. $T = \dfrac{\left(\frac{x_1T_1}{\ell_1} + \frac{x_2T_2}{\ell_2}\right)}{\left(\frac{x_1}{\ell_1} + \frac{x_2}{\ell_2}\right)}$ **42.** $x = \dfrac{(\ell_1 + \ell_2)}{\left(\frac{\ell_1}{x_1} + \frac{\ell_2}{x_2}\right)}$ **43.** 7/2 **44.** (4/3)ω **45.** 10.34 cm

46. 166.3 sec **47.** $\frac{\alpha}{L}\ln\left(\frac{T_1}{T_2}\right)$ **48.** $\frac{K\pi r_1 r_2(\theta_2 - \theta_1)}{L}$ **49. (a)** $\frac{K\pi(R_2^2 - R_1^2)(T_2 - T_1)}{\ell}$ **(b)** $\frac{2\pi K\ell(T_2 - T_1)}{\ln(R_2/R_1)}$

50. $T = \frac{Lms}{KA}\ln\frac{\theta_0 - \theta_1}{\theta_0 - \theta_2}$ **51. (a) 1 : 4 (b) 2.9 : 1** **52. 0.032°C/s and 0.016°C/s**

53. (a) 595 watt/m^2; **(b)** $T_0 \approx 420$ K **54.** 2:1 **55.** 1000 J (C°)$^{-1}$ **56.** $\left(\frac{6}{\pi}\right)^{1/3}$

57. 5800 K **58.** $T\sqrt{\frac{R\sqrt{1-\alpha}}{2D}}$ **59.** $\pi\sigma L^2\left(T_1^4 - T_2^4\right)$ **60. 74 W/m–K**

61. 1.8 W/m–K **62.** 3025 K **63.** 10 min. **64.** 80 kcal/kg **65.** The first one

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EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	3	1	2	4	3	3	2	4	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	4	3	3	1	60	40	20	2	1
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	3	57	3	2	1	8	1	30	2	2
Que.	31	32								
Ans.	3	16								

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	C	A	2	A,B	B	9	A	C	4 (3.99,4.01)
Que.	11	12	13	14	15					
Ans.	2	270	B,C,D	8.32 to 8.34	9					

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Important Notes

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